

ISSN 2394-997X (Print)
ISSN 3107-829X (Online)

Lex Revolution

Periodical Index
Journal of Social & Legal Studies
Volume-XII (1) Jan-Mar 2026

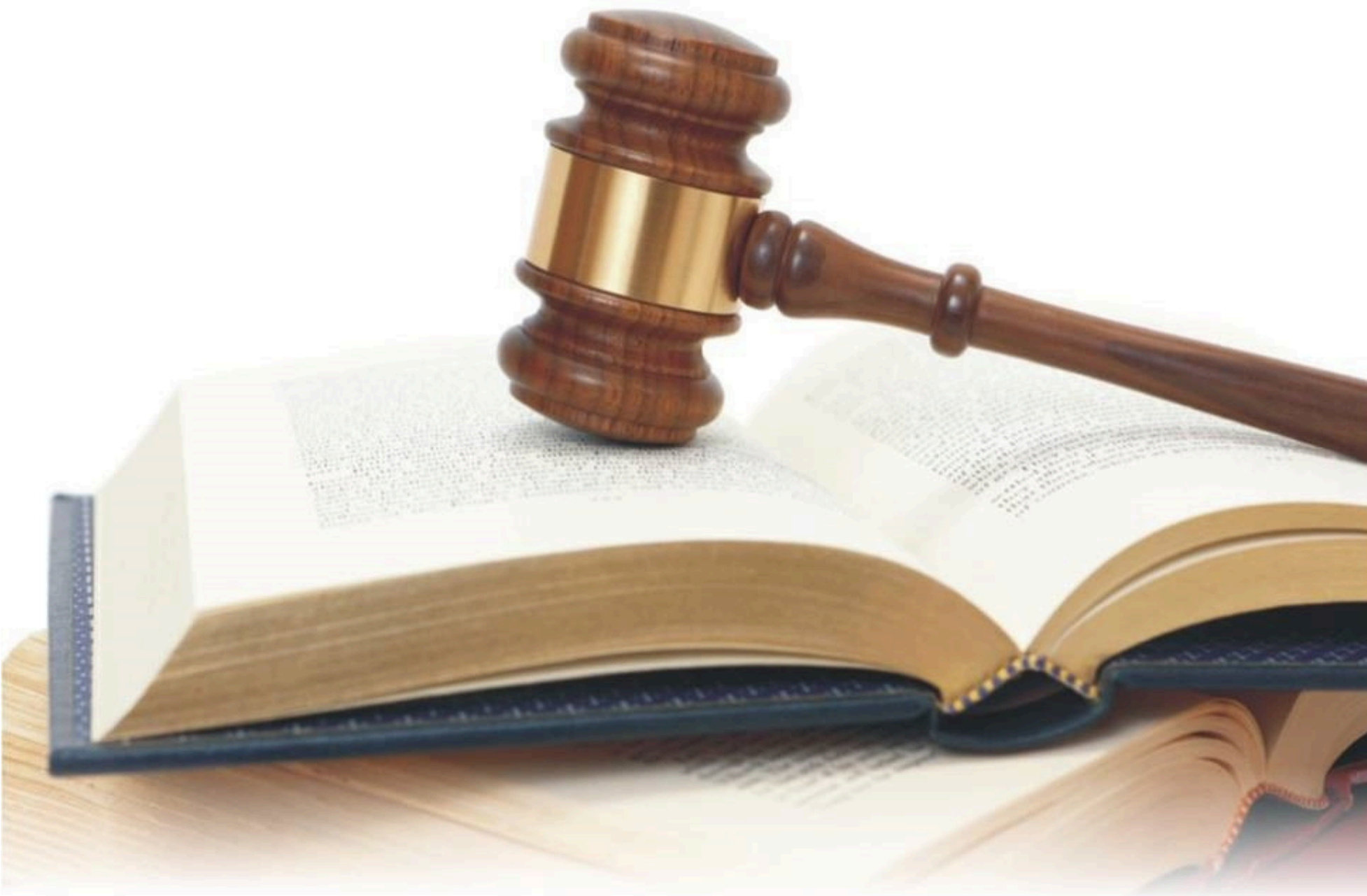


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Volume-XII

Issue-1

Jan-Mar 2026



Published by
Dr. V. B. Pandey
www.lexrevolution.in

“Human trafficking cannot be understood, and still less effectively addressed, through any single lens. To illustrate, a purely criminal justice response, however necessary it may be, leaves untouched the structural vulnerabilities that continuously supply the crime with victims. A purely structural response, however well-intentioned, offers little immediate relief to those already trapped in exploitation and in need of rescue and legal redress. No single framework is sufficient to address the complexity of the phenomenon, and effective anti-trafficking work requires, at its core, a multidisciplinary approach. This is the frame within which this Court wishes to approach and address the issues before it.”

J.B. Pardiwala, J. in

Prajwala v. Union of India

2026 SCC OnLine SC 1053, Para 90

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Lex Revolution

Periodical Indexed Peer-Reviewed Journal of Social & Legal Studies

Quarterly Published Open Accessed Research Journal

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VEDIC SCIENCE AND NATURAL LAW: A COMPARATIVE STUDY WITH MODERN JURISPRUDENCE

- Dr Keshav Jha*

Abstract

The connection between law, morality, and the natural order has been a big deal for people throughout history. Vedic science in ancient India expressed a deep understanding of cosmic order through concepts such as ṛita and dharma, which governed both the universe and human conduct. In Western legal philosophy, the theory of natural law asserts that law derives its legitimacy from universal moral principles intrinsic to nature. This article offers a comparative analysis of Vedic concepts of cosmic order and contemporary natural-law jurisprudence, emphasising their philosophical similarities and distinctions. It also looks at how these traditions have affected goals for education, moral frameworks, and legal awareness. By analysing both systems, the paper argues for a more integrated approach to jurisprudence that harmonises scientific understanding, ethical reasoning, and legal principles in contemporary education and society.

Keywords: *Vedas, Legal Philosophy, Rita, Dharma, Natural Law.*

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INTRODUCTION

Law is not just a set of rules that people in power make; it is also closely linked to moral philosophy and how we understand the world around us. Throughout civilisations, intellectuals have endeavoured to discern universal principles that regulate both the natural world and human conduct. In ancient India, Vedic science provided a comprehensive perspective in which the universe functioned according to an intrinsic order referred to as “rita”. This order manifested in human existence through dharma, which delineated ethical behaviour and social obligation.

Modern jurisprudence, especially the natural law tradition, has evolved predominantly within Western philosophy. Natural law theory posits that specific rights and moral principles are inherent to human nature and can be discerned through rational inquiry. Thinkers such as Aristotle and Thomas Aquinas emphasised that human laws must align with these universal principles to be just.

This article seeks to compare the Vedic conception of natural order with modern natural law jurisprudence. It examines their philosophical foundations, their approaches to ethics and law, and their implications for education. By doing so, it aims to highlight the enduring relevance of these ideas in shaping a just and harmonious society.

VEDIC SCIENCE AND THE CONCEPT OF COSMIC ORDER

Vedic science transcends empirical observation; it embodies a holistic comprehension of reality that synthesises physical, ethical, and spiritual aspects. The idea of “rita,” which means the universal order that governs the cosmos, is at the centre of this way of thinking. The principle that makes sure that natural events happen regularly, like the movement of celestial bodies and the changing of the seasons.

As part of this cosmic system, people are expected to live by this order. This is where the idea of dharma comes from. Dharma takes the abstract idea of rita and turns it into rules for how people should act. It includes duties, rights, ethics, and responsibilities. These things change depending on the situation, but they are always meant to keep things in order.

Unlike rigid legal codes, dharma is dynamic and context sensitive. It requires individuals to exercise judgment and wisdom in determining the right course of action. This flexibility allows dharma to adapt to changing circumstances while remaining rooted in universal principles.

NATURAL LAW IN MODERN JURISPRUDENCE

The theory of natural law in contemporary jurisprudence posits that law is founded on moral principles that are universal and ascertainable through reason. This perspective posits that human-created laws (positive laws) are legitimate solely if they align with superior moral principles.

Aristotle said that justice comes from nature and that laws should help everyone. Later, Thomas Aquinas came up with a more organised theory that said natural law comes from divine reason and can be understood by humans.

Natural law has had an effect on the growth of human rights, constitutional law, and international legal systems in the present day. It stresses values like equality, justice, and fairness, which are thought to be part of what makes people dignified.

However, natural law theory has also faced criticism for its reliance on abstract principles that may be interpreted differently across cultures. Despite this, it remains a significant framework for evaluating the legitimacy of laws and legal systems.

POINTS OF CONVERGENCE

It can be seen that some parallels exist between the Vedic ideas and the theory of natural law:

- **Undeniable Order:** The presence of an undeniable order within the Universe is recognised in both philosophies. In Vedic philosophy, this concept is “rita”, and the moral order within nature forms a similar principle in natural law.
- **Moral Basis of the Law:** It is claimed that law should be based on moral values. Similarly, both dharma and natural law are concerned with ethics such as justice and responsibility.
- **Active Role of Humans:** As opposed to the passive role of people in maintaining order, they are seen as active participants within those systems. People should live according to the universal order.
- **Merging of Law and Ethics:** While in other theories, law can be separated from morality, in Vedic philosophy and natural law, laws are considered as ethical norms.

It becomes clear that people try to discover order despite different cultural backgrounds.

POINTS OF DIVERGENCE

Even though there are many similarities between them, there are also several important distinctions:

- **Origin:** While Vedic knowledge is obtained through inspiration, natural law results from reasoning.
- **Flexibility or Code:** On the one hand, dharma is always flexible and situationally specific, while on the other hand, natural law was used to establish more stringent laws and regulations.
- **Domain:** The science that is connected to Vedas combines spiritual, cosmic, and social perspectives, whereas natural law applies exclusively to humans and their society.
- **Practice:** While dharma is an intrinsic part of everyday human activities, natural law is more applicable to the field of law and politics.

IMPLICATIONS FOR EDUCATION

Education plays an important part in developing knowledge about the universality of principles, both in the Vedic tradition and natural law theory. Within the Vedic system, education included not only the development of intellectual capacities but also the training of a person to perceive his dharma and behave appropriately.

Natural law theory also emphasises the necessity of education for a person to think critically and morally. It teaches people to oppose unjust laws and support the values of universal justice.

Modern trends in education are oriented towards the realisation of the necessity of creating value-based educational systems capable of producing specialists and responsible citizens.

Thus, one can apply ideas of both approaches to create a modern system that includes not only the learning of knowledge but also the perception of ethics, responsibility towards the environment, and other important factors.

RELEVANCE IN THE MODERN WORLD

Today's world is complicated in its nature, and problems like pollution, discrimination, and ethical issues necessitate a better understanding of the connection between laws and morality. In this regard, the idea of coexistence with nature inherent in Vedic culture and the universal rights philosophy inherent in natural law can serve as helpful approaches.

It should be concluded that a combination of both approaches can help create more advanced and responsible legal systems that will contribute to sustainable development and well-being.

CONCLUSION

As shown above, the examination of Vedic tradition in comparison with the modern ideas of natural law and its application to jurisprudence demonstrates remarkable parallels between these philosophical systems. In other words, both approaches share the idea that laws should have ethical roots to be justified and effective.

Despite differences in their sources, methodology, and scope of influence, both of the approaches discussed above underline harmony, justice, and responsibility as important universal principles. Thus, the philosophies of rita and dharma presented in the Vedic culture and natural law theory provide us with a universal worldview.

In relation to today's education and jurisprudence, however, this integration can result in a balanced and productive methodology. Through moral awareness, intellectual development, and increased responsibility, education will enable individuals to make their contributions to a just and peaceful society.

In essence, the lasting importance of these cultures is their capacity to steer humankind to greater knowledge about law, morality, and the universe. The integration of these cultures is a way through which the current generation can address its difficulties while looking forward to an enlightened future.

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VEDIC SCIENCE AND ITS RELEVANCE IN MODERN EDUCATION

- Dr Keshav Jha*

Abstract

The Vedas constitute an intricate aspect of the Sanskrit literature. They include all the ideas, concepts, philosophy, history, sciences, art, and literature that are present in Sanskrit literature. The Vedic science constitutes a holistic way of life; as opposed to the alien way of education, it does not incorporate in itself the intellectual, ethical, spiritual, and practical aspects. The Vedic science constitutes a unified science or system. It is a science of scientists. It is a science of sciences. The result of Vedic science lies in completing and perfecting the man, which is accomplished by education and life experiences. The objective of this paper is to discuss Vedic science and analyse the significance of Vedic science in contemporary education.

Keywords: Vedas, Education, Humanities, Arts, Holistic Development.

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INTRODUCTION

Education is one of the most significant pillars of human civilisation. The modern education system has made huge contributions to science and technology, as well as to integrating the world. However, the morals, emotional quotient and holistic growth are sidelined, and the students are made to meet the expectations of society by completing the process of education. Students are alienated from their natural surroundings and do not know themselves. Duality is created in an individual. The whole focus of the modern education system has been on materialism, while the real essence of human beings has been completely overlooked. Vedantic science is not a concept restricted to religious and ritualistic science alone. It involves philosophy, mathematics, astronomy, medicine, psychology and several other forms of science. The ancient Vedic education system did not involve just knowledge but experience, wisdom and self-realisation as well. In this write-up, I have tried to analyse the significance of Vedic science in today's education system.

HOLISTIC APPROACH TO EDUCATION

Education is one of the most crucial elements of Vedic science. The Vedic education system is centred on the physical development of the body, mind and spirit. Contrastingly, contemporary education centres on the mental development of the students. The contribution of Vedic education to education systems includes the formation of a holistic approach to education. It is concerned with various issues relating to the personal development of the students, thus creating a significant area in education. Another major contribution of the Vedic education system to the education field includes the formation of a physical, mental and spiritual education model.

EMPHASIS ON ETHICAL AND MORAL VALUES

Value-based education that carries the most value. Among the modern education systems, an absence of values like honesty is felt greatly. While technical skills might be plentiful, values such as morality and ethics have been sidelined in many cases. In the Vedas, great importance was given to the practice of honesty, non-violence, discipline and accountability. These values were not taught in isolation; they formed a significant part of daily life practices. With value-based education based on the principles of Vedic science, one can learn the skills necessary to deal with the complexities of moral issues in real life.

RELATIONSHIP BETWEEN TEACHER AND STUDENT

The Vedic model of education emphasised that there had to be a good rapport between the teacher and the learner. Education was not confined to the classroom. The relationship between the teacher and the student

The Vedic model of education emphasised that there must be rapport between the teacher and the learner. Education was not restricted to classrooms but extended to all aspects of day-to-day living.

The personalised method of teaching enabled the teacher to know his or her learners better and thus make the learning process more efficient and effective.

Modern education systems face many difficulties due to overcrowded classes and the uniformity of instruction methods. Reviving certain features of the mentor-student relationship will encourage students to participate more actively, be more motivated, and develop socially.

KNOWLEDGE AND INTEGRATION ACROSS DISCIPLINES

The field of knowledge in Vedic science is not separated into clear-cut disciplines. Instead, they consider the knowledge to be interconnected and interdependent. The study of astronomy is interlinked with mathematics and philosophy, while medicine is interlinked with ecology and spirituality.

While today's education system seems to adopt an interdisciplinary approach in the classroom, there is still an inclination to classify them into various disciplines, which makes it difficult for students to relate the knowledge they learned to real-life situations.

Utilising concepts from Vedic science allows students to acquire greater knowledge about a particular field of study, which would encourage critical and creative thinking.

MENTAL HEALTH AND AWARENESS

Student mental well-being is becoming one of the major concerns within today's schools. Conditions such as stress, anxiety, and depression are triggered by excessive academic pressure, competitiveness, and a lack of friends.

In regard to Vedic science, students will be provided with valuable methods for maintaining their psychological well-being, such as meditation, breath control exercises, and self-awareness techniques. As a result, they will manage to increase their concentration, control their emotions, and, therefore, maintain their psychological health.

The introduction of such activities in a school setting will have a positive impact on student performance and psychological well-being.

Experiential Learning

Learning through experience was the basis of Vedic education. The emphasis was placed on observation, asking questions, and applying theoretical knowledge to practice.

This approach corresponds to the contemporary ideas about active learning.

AWARENESS OF THE ENVIRONMENT AND SUSTAINABILITY

The importance of harmony between human beings and the natural world is emphasised in Vedic science. Natural objects were considered sacred, and it was believed that living in harmony with the environment was an essential aspect of life.

These values are extremely relevant in our time, when environmental issues have gained significant relevance. Educators could take advantage of Vedic ecological knowledge and make it part of their curriculum.

It would increase awareness about environmental responsibility among students by highlighting the significance of conserving natural resources.

FOCUS ON SELF-DISCOVERY AND PURPOSE

What sets Vedic education apart from other types of education is that it is self-actualising in nature. It was believed that education would help individuals know themselves better.

In contemporary times, people put more importance on how well education enables them to progress in their careers and earn money as opposed to developing into well-rounded individuals.

In order for education to contribute to the overall development of individuals, they need to be able to discover meaning and direction in their lives through education.

LANGUAGE, MEMORY, AND BRAIN DEVELOPMENT

The ancient traditions of Vedic learning emphasise oral learning, memory and speaking properly. Methods of chanting and repetition were among those methods that helped in sharpening memory and thinking skills.

Modern scientific research supports the use of similar methods as useful means for helping people with concentration, memory and the development of their brains.

Inclusion of such methods in contemporary education can be beneficial for people as they will help in developing their cognitive skills and learning more effectively.

The incorporation of Vedic science into the contemporary education system is justified, as there are several issues that arise when doing so. Some examples of challenges related to implementing this strong scientific knowledge include cultural differences, the requirement of evidence, misrepresentation and superficial implementation of Vedic science, among others. Therefore, it is necessary to explore ways in which Vedic Science can be implemented in your education system without being too rigid and considering the universal aspects of Vedic Science as well. Moreover, all educators must collaborate in formulating an education policy that is balanced.

CONCLUSION

The Vedic Sciences provide a myriad of holistic approaches to counteract the problems inherent in the current education system. Taking advantage of the teachings of Vedic Science, focusing on ethics, enhancing mental wellness, creating connections within the sciences, and facilitating learning, as well as offering teachers, learners, and policy-makers endless opportunities to construct a holistic educational experience.

Education cannot be considered just an informational conduit. Education must train learners to become individuals armed with knowledge, wisdom, empathy, and societal awareness in the rapidly evolving era fuelled by technology. Integrating the principles of Vedic Science in today's education system does not amount to replacing the existing curriculum. Rather, it enhances the current educational experience with ancient knowledge. Merging humanity's long-gone knowledge with the contemporary education system creates an education system that caters to humanity's needs at the individual, communal, societal, and spiritual levels. Constructing such an education system would ultimately ensure a cooperative and sustainable future.

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BRIDGING PROMISE AND REALITY: EVALUATING THE RIGHT TO EDUCATION ACT IN TODAY'S INDIA

- Dr Avnish Singh* & Sagarika Singh**

Abstract

A key rights-based intervention, the Right of Children to Free and Compulsory Education Act, 2009 (RTE Act), under the provisions of Article 21A of the Constitution, is an attempt to ensure universal elementary education in India. However, over fifteen years after its promulgation, the effectiveness of the operational facet of the Act continues to be the point of dispute among scholars. It is through this exposition that the critical socio-legal analysis of the RTE Act in modern times is provided, which focuses on empirical realities that are evident at the grassroots level. Even though the Act has produced surface improvements in enrolment, access, and gender parity, the improvements have been unsuccessful in converting into learning outcomes. The paper, relying on governmental reports, judicial proclamations and empirical studies, including the Annual Status of Education Report (ASER) archives the numerous challenges still faced, including insufficient infrastructure, lack of teachers, unequal delivery, socio-economic and cultural obstacles, migration trends, discrimination and the digital divide. COVID-19 also made evident the systemic lack of digital preparedness. In the article, it is argued that the future success of the RTE Act depends on the paradigm shift between the compliance-based framework towards the learning outcomes and the inclusivity-based framework. This shift would require greater teacher capacity, fair infrastructure development, strong governance systems, and context-directed policy reforms, which would bring to fruition the constitutional promise of quality education for every child.

Keywords: Education, RTE, Educational Inequality, Fundamental Rights, Article 21-A

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INTRODUCTION

The Right to Education Act, 2009¹, is one of such legislations in India that has been the most ambitious in playing a role in changing the education scenario in the country. Its implementation was in response to the constitutional right of education as an essential right under Article 21-A of the Constitution, and the RTE Act ensures the right of free and compulsory education among children between the ages of 6 and 14 years. Over 25 years of implementation, the Act remains a debatable issue among policymakers, scholars, civil society stakeholders, and grassroots practitioners. The main question that lingers is whether the promises of the Act have been made into reality on the ground level.

The article provides a multi-dimensional approach to investigating the effectiveness of the RTE Act in the current times. It discusses its success, systemic obstacles, challenges on the ground, and the degree to which the Act has been effective in meeting equitable, inclusive and quality education. Since India is trying to harmonize its educational goals with the Sustainable Development Goals (especially SDG 4: Quality Education)², it is highly topical and needed to analyze the practical effect of the RTE Act.³

The real sense of Indian education of present day is a different story on the ground level. Firstly, enrolment rates have increased, access has been enhanced and awareness created. Conversely, poor infrastructure, shortage of teachers, crisis in learning outcomes, socio-economic factors and uneven mechanisms of implementation are some of the problems that hinder the desired results of the Act. The COVID-19 pandemic also exacerbated and increased the digital divide, which increased inequalities among children who belong to various socio-economic classes.⁴

The article, written in thematic sections, tries to fill the gap between the intent and the experience of the legislation on the ground. It is a critical analysis of the provisions, outcomes and limitations of the Act in relation to empirical data, government reports, scholarly literature and field-based observations. The aim is to deliver a subtle, detailed evaluation, which will capture the modern-day realities of the education system in India.

¹ Right of Children to Free and Compulsory Education Act, No. 35 of 2009, INDIA CODE (2009).

² Government of India, NITI Aayog, *SDG India Index & Dashboard 2023-24*, especially SDG 4 indicators.

³ Ministry of Human Resource Development, *Right of Children to Free and Compulsory Education Act, 2009: Status Report* (2014).

⁴ Mehendale, Archana; Mukhopadhyay, Rahul & Namala, Annie, "Right to Education: Challenges of Policy Design and Implementation," 49 (9) *Economic and Political Weekly* 23 (2014).

HISTORICAL BACKGROUND AND EVOLUTION OF THE RIGHT TO EDUCATION

The idea of education as a human right in India did not take place overnight, but was a decades-long struggle, policy development, and judicial action. Most of the Indian education system, before the RTE Act, was managed by an assortment of policies such as the National Policy on Education (1986), Sarva Shiksha Abhiyan (SSA), and a variety of state-level policies. These programmes made access significantly better but were unable to provide uniformity, equity, and enforceable rights.

The historical decision of *Mobini Jain v. State of Karnataka*⁵ by the Supreme Court. The broad framework was provided in this case, which proclaimed the right to education implicit in the right to life in Article 21. This was restated in *Unnikrishnan v. State of Andhra Pradesh*⁶, which defined the State as having a responsibility of providing free education to children up to 14 years of age. It was finally in 2002 with the constitutional amendment that Article 21A was added, making statutory enforcement possible in the form of the RTE Act.

This solid constitutional foundation notwithstanding, there is still a lot of breach between policy vision and ground reality. The access to education by children is still influenced by the socio-economic inequalities, caste and gender discrimination, and urban-rural divide that occur historically. This historical course is necessary to assess the contemporary applicability of the Act.

Indian socio-political ambitions are also manifested in the development of the right to education. Education has been traditionally regarded as a means of nation-building, social transformation and empowerment of the marginalised groups. However, for a large portion of independent Indian history, educational policy was disjointed and underinvested. The Directive Principles under the Constitution, especially Article 45, did envisage free and compulsory education to all children, but due to the unenforceability of the same, the process was slow and uneven.

During the 1980s and 1990s, the increased activist engagement of civil society, scholarly studies and international commitments, including the Education for All (EFA) movement, started to bring education to the centre of policy debate. The emergence of grassroots movements, such as the Lok Jumbish Project, the District Primary Education Programme (DPEP) and movements

⁵ (1992) 3 SCC 666. See, Animesh Kumar, “Evolution of Right to Education in India” 2 (1) *Law and Society: A New Challenge* 27 (2015)

⁶ (1993) 1 SCC 645. See, Animesh Kumar, “Right to Education in India: The Role of Judiciary” 4 (2) *Indian Journal of Socio Legal Studies* 39 (2015)

organised by such organisations as the Right to Education Forum, brought to the fore the fact that the rights-based approach was urgent.

Moreover, India is a member of such international organisations as the UN Convention on the Rights of the Child 1989, and it imposed pressure on the state to adjust domestic legislation to the international standards, especially those that relied on universal education. The progressive pressure of case law, political discussions, grassroots movements, and global commitments provided an opportune condition towards the process of passing the RTE Act in 2009.

However, despite the enactment of the Act, the history of historical inequalities still hangs high. The history of the right to education is not just a legal process but also an example of a sophisticated process of democratising the right to knowledge and breaking the system of social stratification in the country. This is the multi-layered history to understand the context of how the Act is performing within modern times.

KEY PROVISIONS OF THE RTE ACT AND THEIR INTENDED OBJECTIVES

According to the provisions of the RTE Act, a few transformation provisions were formed to restructure the educational ecosystem. They involve free and compulsory education, neighbourhood schools, a ban on capitation fees and screening processes, teacher-pupil ratios, infrastructural norms, inclusive education policies, the 25% quota of disadvantaged children in the private unaided schools, and School Management Committees to decentralise the management of schools.⁷

These provisions were aimed at not only increasing access but also quality and accountability. As an example, the required teacher training and the ongoing development of the qualification requirements were to professionalise teaching. School infrastructure norms, which included classrooms, separate boy and girl toilets, playgrounds and drinking water, were instrumental in developing a conducive school environment.⁸

Nonetheless, inconsistencies have dogged the implementation process. Some states were quick to adapt, whereas others were slow because of financial limitations, bureaucratic delays or political apathy. The outcome has been a patchwork of development that is seen in urban centres but not in the remote and disadvantaged areas.

⁷ Ministry of Human Resource Development, 'The Right of Children to Free and Compulsory Education Rules, 2010.

⁸ National Council for Educational Research and Training (NCERT), 'Teachers' Manual on Continuous and Comprehensive Evaluation (2010).

The other key guarantee of the Act is the no-detention policy that guarantees that pupils are not detained or expelled until the end of elementary schooling. This was to alleviate psychological pressure and promote retention. Although this was well-intended, this provision has been reproached because it has led to a drop in learning levels, and states have been allowed to reintroduce some exams under some conditions. However, the provision can be seen to be more inclusive of child-friendly education than the Act is.

Curriculum reform is also a focus of the Act, with the requirement that the curriculum should be changed towards learning activities that are child-focused and aimed at holistic development as opposed to rote learning. This is supported by the philosophies of education advocated by such philosophers as Mahatma Gandhi and Rabindranath Tagore, who saw schools as a place where people can be creative, think critically, and develop democratic values.

Also, it is necessary to create School Management Committees (SMCs) consisting of 75 percent parents to democratise the school and improve the involvement of the community. SMCs are viewed as watchdogs that should control the school operation, fund use and enforcement of norms. The truth of the matter is that most of these committees are not trained, not aware or not empowered with the real decision making, which limits their efficacy.

Reservation of 25% in Section 12(1)(c) is one of the most ambitious and controversial sections of the Act. It strives to ensure social integration by allowing children in the economically disadvantaged groups to enjoy private education. Although the policy has seen thousands of children join the elite institutions, there are such problems as late reimbursement, bureaucracy, and social discrimination. However, this provision indicates a daring effort to shatter socio-economic obstructions in the stratified Indian system of education.⁹

Also, the Act requires that all schools be recognised according to the infrastructure and quality standards. The unrecognised schools had no option but to upgrade or close their doors in a bid to do away with poor schools. Nevertheless, this clause has equally been criticised for having a disproportionate impact on low-cost private schools operating in low-income communities.¹⁰

The RTE Act, in principle, provides a detailed guideline on universal, equitable and quality elementary education. However, the difference between intent and practical application at the

⁹ RTE Act, 2009, s. 12(1)(c)

¹⁰ Prachi Srivastava, "Low-fee Private Schooling and the RTE Act: A Critical Examination," 47(23) *Economic and Political Weekly* 51 (2012).

level of the legislative process emphasises the necessity of more powerful enforcement tools, increased financial support, and political follow-up.

ACHIEVEMENTS AND SUCCESS INDICATORS OF THE RTE ACT

Still, the RTE Act has introduced some major changes in the educational situation in India despite the existing difficulties. Primary school enrolment has always been over 95 percent, dropouts have also reduced, and gender parity has also improved in some states. Such schemes as mid-day meals, free textbooks and uniforms have helped in this development.

The 25% quota in Section 12(1)(c) has afforded the children in the financially disadvantaged groups to learn with their counterparts in the private schools. Although the implementation has been different, most states have documented effective integration and enhanced socialisation among children. The other success is in the fact that there is greater accountability within the education system. Schools have become more conscious of their obligations, parents are more knowledgeable about their rights, and the civil society organisations are becoming increasingly engaged in monitoring.¹¹

These accomplishments should, however, be approached cautiously. Growth in enrolment does not necessarily result in better learning outcomes. In the Annual Status of Education Report (ASER), it is recurrently noted that children in most grades can neither read nor do basic arithmetic at the grade level, and this is a reflection of a serious learning crisis.

The RTE Act has also enhanced more inclusivity and social justice on top of enrolment and access, since children of marginalised communities and special categories have to be enrolled. Children of Scheduled Castes, Scheduled Tribes, Migrant families, and children with disabilities have enjoyed the provisions that are tailor-made to solve their barriers in education. A number of states have implemented some specific interventions, including bridge courses, seasonal hostels, and inclusive classrooms, reinforcing the equity-based view of the Act. The other significant success is the improvement of school infrastructure in the country. Thousands of schools were improved with the implementation of the RTE norms, through the building of classrooms, toilets, drinking water systems, and school boundary walls. These gains, albeit uneven, have aided the establishment of safer and child-friendly learning environments.

¹¹ Supra note 9.

Evidence-based planning is also shifting towards technology-based systems used to trace infrastructure gaps and teacher deployment, adopted by many states.¹²

The Act has also enabled decentralised governance structures, especially School Management Committees (SMCs). Transparency in the school operation has been enhanced by the active participation of parents particularly mothers. SMCs have been used in checking the attendance of teachers, the delivery of entitlements in a timely manner, and in handling school development plans. The development of such participatory structures is a major step to community-led education, although the effectiveness of these structures varies. Also, the RTE Act has encouraged state and national policy innovation. Programmes such as Learning Enhancement Programmes (LEP), remedial programs, basic literacy and numeracy motivational programmes and online instructional tools have become increasingly popular as governments seek to conform to the quality provisions of the Act. Kerala, Himachal Pradesh, and Tamil Nadu are the examples of the states who have shown significant improvements with great governance and investment in teacher training.¹³

However, the main problem is the discrepancy between access and learning. Although the Act has managed to take children to school, it has not been able to make them learn effectively once they are in school. This tension remains constant between quality and quantity and underscores the necessity to begin once again to consider pedagogical changes, the capacity of teachers, and systems of assessment. The success of the Act, as big as it is, should thus be viewed as the ground to further enhancement instead of an end to the educational process in India.

PERSISTENT CHALLENGES: GROUND-LEVEL REALITIES

The absence of consistency in implementation is one of the fundamental issues of the RTE Act. The rural and tribal areas still experience chronic teacher shortages, lack of development, multi-grade classes and high absenteeism. A high number of schools fail to comply with the infrastructural standards set forth by the Act, and a number of states have tried to request extensions a number of times.

This is complicated by socio-economic realities. Children of disadvantaged groups are also compelled to participate in labour, household chores, or take care of their siblings, which

¹² Nivedita Menon, "Accountability and Community Participation in School Governance," 52(32) *Economic & Political Weekly* 23 (2017).

¹³ R. Govinda, & Madhumita Bandyopadhyay, "Access, Participation and Progress in Elementary Education," 45(36) *Economic & Political Weekly* 47(2010).

restricts their chances of going to school on a regular basis. Females, especially girls, are exposed to numerous obstacles such as security issues, child marriage, difficulties in menstruation and domestic roles. The issue of caste-based discrimination in schools still continues in most schools and can be seen in the form of separation during seating, different treatment by the teachers and in group activities. Equally, children with disabilities still experience challenges because of poor special educators, the absence of available facilities, and poor awareness.¹⁴

The existence of the digital divide led to an additional degree of inequality during the pandemic. Millions of children were unable to get access to smartphones, stable internet, or digital learning devices, virtually kicking them out of the education system. The framework of the RTE Act failed to foresee such a crisis, and thus, there was a need to modernise. The other chronic problem is associated with the teacher quality and training. The pedagogical expertise of teachers is usually not as high as needed to provide child-centred, competency-based education even in those settings where teachers can be found. Teachers have a non-academic workload to complete, including working in elections, taking census surveys, and administering the administration, which makes them spend even less time in the classroom. States that have a high population of para-teachers or contract teachers have an added challenge in upholding standards of teaching because they tend to be volatile, poorly paid, and do not have opportunities for professional development.¹⁵

The problem of school governance can also influence proper implementation. Although School Management Committees are supposed to democratise the decision-making process, they operate in many different ways. Formation of committees and their representation within certain groups, particularly the marginalised groups, does not create awareness in committee members, and some members are afraid to take action against school authorities. The independence that is desired by these committees is undermined by political interference and bureaucracy in releasing funds. Also, seasonal work and migration do not allow continuity in education. The families that were involved in either brick kilns, building work, farming, or informal labour tend to change their residence regularly, which leads to long-term absenteeism or dropout. Special

¹⁴ Geetha B. Nambissan, "Equity in Education? Schooling of Dalit Children in India," 38 (44) *Economic & Political Weekly* 4581 (2003)

¹⁵ Sukhadeo Thorat & Katherine Newman, *Blocked by Caste: Economic Discrimination in Modern India* (Oxford University Press 2010).

considerations of the migrant children are mentioned in the RTE Act, although there is still inadequate implementation of the same.¹⁶

Another challenge at the ground level is language barriers. Language minorities and tribal belt children usually have difficulties with instructions being taught in languages they do not know well. Although the Act is in favour of mother-tongue instruction, most states do not have resources or materials to make this work. Also the presence of socio-cultural norms which are deep-rooted restricts the involvement of both girls, children with disabilities, and lower caste. Most of the schools already do not have safe transport, boundary wall, proper toilets, or a system of dealing with harassment and discrimination. Those problems do not only decrease the attendance level and also interfere with the sense of belonging and dignity of children in the school setting¹⁷.

All these unresolved problems indicate that the success of the RTE Act does not only lie in the law since it is imperative to consider the more social, economic, and cultural realities that influence how children acquire an education.

TEACHER-RELATED ISSUES: THE BACKBONE OF THE EDUCATION SYSTEM

The most fundamental part of the education system is teachers, and they still happen to be one of the least considered elements in the implementation of RTE. Teacher shortage is an acute problem in many states, which causes overwork and deterioration of teaching standards. Contractual teachers who receive significantly lower wages than their permanent colleagues have low job security, less training, and lack motivation.

Absenteeism of teachers is a significant problem, especially in rural schools. In addition, teachers are often given non-academic tasks like election work, census survey and administration of welfare schemes that limit the time they can spend teaching. The quality of training programmes is very different, although the Act provides regular training. According to reports of many teachers, training is, in theory, irrelevant to classroom realities and not effective in helping teachers acquire new pedagogical skills. Such issues undermine the efficacy of the Act and emphasise the necessity of powerful changes among teachers.¹⁸

¹⁶ Smita, "Distress Seasonal Migration and Its Impact on Children's Education," 44(4) *Economic & Political Weekly* 59 (2009)

¹⁷ Supra note 9 at ss. 3, 8, 9 & 12

¹⁸ M. Kremer *et. al.*, "Teacher Absence in India: A Snapshot," 20(2) *Journal of the European Economic Association* 658 (2005)

The problem has even greater depth considering the regional differences in the distribution of the teachers. The highest pupil to teacher ratios are in the states of Uttar Pradesh, Bihar, Jharkhand and Rajasthan, which are much higher than the norms provided by the RTE and other southern and northeastern states are relatively good. This disproportion is usually caused due to lack of proper planning, tardiness of recruiting procedures and inefficiency in administration. In the distant tribal regions, teachers do not want to get posted because of the poor facilities, commuting distance, and incentives. Therefore, children in such areas are the ultimate victims of structural neglect.¹⁹

The other issue is the current disparity between the qualification of the teachers and the classroom requirements. Although the RTE Act reflects professional qualification, most teachers have not been well prepared in basic literacy and numeracy instructions, inclusive education, or multilingual education. National achievement survey (NAS) reports and ASER reports all bring attention to the fact that most classroom practices are based on rote learning, lack of student engagement and the use of old approaches to teaching. This erodes the pedagogical vision of the Act which is child-centric.

Moreover, poor accountability is caused by the lack of effective teacher evaluation system and feedback. The current evaluation systems are more based on administrative conformity rather than pedagogical efficiency. Participatory feedback mechanisms, peer based learning framework and continuous mentoring that are necessary to support growth in teachers are feeble or non-existent in most states. Education models like Finland, Singapore and South Korea that have been successful invest a lot in teacher mentorship which India has not yet to scale up.²⁰

In addition to this the educators tend to experience emotional pressure and burnout. The size of classes, the necessity to finish the syllabus, working with the administration and the absence of support staff adds to their workload. Instructors in unsafe or under-resourced settings become more stressed and this impacts their morale and performance as teachers.

The absence of technological preparedness among teachers is one of the most underestimated fields. The digital gap was revealed during the COVID-19 pandemic. However, some of the teachers adjusted fast, others could not cope with the digital platform, create online materials, or interact with the students online. Although the government has been encouraging digital

¹⁹ V. Ramachandran, M. Pal, S. Jain & S. Shekar, *Teachers in the Indian Education System: How We Manage the Profession* (Azim Premji University 2018).

²⁰ National Council for Educational Research and Training (NCERT), *Teachers' Manual on Continuous and Comprehensive Evaluation* (2010).

education, there has been no adequate revising of the teacher training modules in order to develop a technological capacity. Also, the issue of teacher accountability is usually politicised. In some states, teacher unions determine the process of transfer, slow down disciplinary action and oppose reforms like biometric attendance systems. Although unions have a great role in ensuring the rights of teachers, they at times make it very difficult to keep the teaching profession transparent and efficient.²¹

At the end the teachers are also demotivated by the absence of social recognition. The women teachers in most communities are faced with safety issues, long commuting times, and unprofessional development. In the absence of a friendly ecosystem, teachers will find it hard to provide the quality education they had envisaged in the RTE Act.

On the whole, these interrelated issues demonstrate that meaningful implementation of the Act could not be achieved without improvement of teacher capacity, morale and working conditions. The improvement of the education system needs to be based on the principle of not only hiring more teachers but also investing in their further development, well-being, and empowerment. It is only at this point that the RTE Act is able to have a transformative impact as was intended.

INFRASTRUCTURE GAPS: A PERSISTENT STRUCTURAL CHALLENGE

The infrastructure plays a major role in the learning experience of a child, and most of the schools in India still do not have basic facilities. A number of government reports point out that most schools lack proper toilets, clean drinking water, adequate classrooms, ramps accommodating the disabled children, libraries, internet and walls. Lack of these facilities directly affects attendance and learning. An example is that the absence of toilets among girls leads to absenteeism in adolescent girls, and the absence of electricity also obstructs the use of electronic tools and contemporary teaching techniques.²²

Despite the fact that clear norms are stipulated by the RTE Act, there is not much enforcement. This crisis is caused by budgetary limits, bureaucratic and inappropriate use of funds. Infrastructure strengthening cannot be achieved through the provision of funds only but also through accountability and involvement of the community. The issue of poor infrastructure is even worse in the rural and tribal areas and the hilly landscapes whereby the terrain is very difficult, less connected and transportation is less developed thus it is much harder to build and

²¹ NITI Aayog, School Education Quality Index (SEQI): Teacher Performance and Governance Indicators (2021)

²² Ministry of Education, Government of India, Unified District Information System for Education Plus (UDISE+) Report 2021-22.

maintain school buildings. Most of the schools are run in temporary buildings or run down buildings where children are exposed to risks of safety. Extreme weather conditions, landslides, and seasonal flooding usually make the situation worse resulting in constant school closures that hamper continuity in learning.²³

Also, the need to have inclusive infrastructure has persisted in marginalizing children with disabilities. Although the Act requires barrier-free access, there is a sharp contrast on the ground as things are. Classrooms, toilets, playgrounds, and teaching materials are inaccessible even in the places where they have ramps. The lack of support tools and resource teacher trainers also contributes to the inability of children with special needs to participate fruitfully.

Another issue of concern is digital infrastructure gap. Most of the schools have no computers, projectors or good internet access, even though the national initiatives have been encouraging learning based on ICT. The COVID-19 pandemic put this gap into a very clear perspective: schools that lacked a digital infrastructure failed to implement a blended or online learning model, making millions of children victims of learning loss. This revealed the shortcomings of the RTE Act which was formulated when digital education was not the focus of schooling.²⁴

The second aspect of infrastructural insufficiency is that sanitation and hygiene facilities of most government schools are poor. No proper toilets, lack of running water and bad maintenance conditions have a negative impact on the health and dignity of students. This is especially on girls, who might miss school during their menstrual days because they are not given any privacy or even a basic hygiene product. Finally the development of infrastructure is often brought to a standstill by delays in the allocation of funds and a lack of coordination between the central, state, and local governments. Bureaucratic delays, corruption and absence of technical supervision mean poor construction or incomplete project despite the sanction of funds.²⁵

In general, industrial solutions to infrastructure gaps must be multi-layered, which involves making specific investments, open monitoring, capacity building of the local authorities, and active participation of the communities. The overall objectives of the RTE Act cannot be achieved without good, safe, and inclusive infrastructure.

LEARNING OUTCOMES CRISIS: BEYOND ENROLMENT

²³ National Institute of Educational Planning and Administration (NIEPA), *Elementary Education in India: Progress Towards RTE Norms* (2020).

²⁴ Jean Dreze & Amartya Sen, *An Uncertain Glory: India and Its Contradictions* (Princeton University Press 2013).

²⁵ Ministry of Jal Shakti, Government of India, *Swachh Vidyalaya Initiative: National Assessment Report* (2019).

The increasing disparity between enrolment and actual learning is one of the most serious issues that are provided in the analysis on the ground level. Children tend to advance in grades in a manner that they do not get the basic skills. This grade slippage leaves accumulation of deficits, which is detrimental to future learning. The students, who cannot read fluently and do the basic arithmetical procedures when they enter the upper primary level are at a great disadvantage as they lose confidence and lose interest in the learning process all together.²⁶

The policy of no-detention that was based on progressive principles was criticized as having contributed to low learning results. Although the policy was supposed to ensure stress-free learning environment and a decrease in the rate of drop outs, most educators disapproved that the policy was indirectly lowering accountability among the teachers and students. In automatic promotion, schools used to pay much attention to finishing the syllabus and neglecting conceptual knowledge. Even though the policy has since been revised to permit the use of performance during detention on basis of re-examination attempts, the fundamental issue of it all remains, which is the necessity of changing the mode of learning to competence based learning.²⁷

The regular reports by ASER indicate that very high proportions of children in Grade 5 are unable to read Grade 2-level text. According to this learning crisis, the implementation of the Act has been distorted so that it has focused more on admission and not on quality of teaching or pedagogy. The focus on the infrastructure and enrolment statistics has at times eclipsed what has equally been a significant task of refining classroom activities, evaluation systems, and the learning conditions. Consequently, children who are on paper in school may not be getting appropriate education.

The other difficulty is the curriculum which has often been accused of being thick, examination based and unrelated to the actual lives that students live. Most learning environments continue to use rote learning and have few alternatives in terms of critical thinking, creativity and interpersonal learning. The lack of connection is also exacerbated by massive classes, inadequate teaching help, and the absence of individualised teaching of slow learners.

Systemic reforms are necessary to handle this problem. Improved assessments based on competences and not on memory, continuous training of teachers in the new style of pedagogy, and curriculum changes based on experiential learning can contribute greatly to better results.

²⁶ Jean Dreze & Amartya Sen, *An Uncertain Glory: India and Its Contradictions* (Princeton University Press 2013).

²⁷ National Council of Educational Research and Training (NCERT), *Learning Outcomes at the Elementary Stage* (2017).

Technology may also transform: digital learning tools and adaptive learning environments as well as smart classrooms can offer ways to personalise the learning process and minimise gaps. Nonetheless, the implementation of such tools should be regarded with care, and these tools should be made to be equal so that the rural and marginalized communities can have equal access. Community involvement and school supervision as well as academic support programmes such as remedial classes may be used in bridging the gap in learning as well. Finally, enhancing the learning outcomes should be the main target of the RTE implementation in the next decade.

SOCIO-ECONOMIC AND CULTURAL BARRIERS: THE INVISIBLE OBSTACLES

Education is not done in a vacuum. The realities of socio-economic enable children to see, acquire, and utilize schooling. One of the largest hindrances to education is poverty. Poor families are faced with the challenge of providing and satisfying subsistence needs rather than providing and satisfying future education results. Children, particularly boys, in most houses are supposed to support their families by working, and girls often have to cope with domestic duties. This financial necessity drives children into inconsistent school attendance or total school dropouts. Although schooling may formally be free, other costs like uniforms, transport, school books and even private tutoring still weigh down low income families and detract effective participation.²⁸

Gender inequality has an impact on education as well. Although gender parity in the primary level has improved, the dropout rates of adolescent girls are still high since some girls are marrying at very young ages, they do not feel safe and their social norms do not favor the education of girls. Girls in most rural regions have mobility limitations preventing their access of the schools that are far away. Stigma associated with menstruation as well as lack of good sanitation facilities also impact attendance. These deeply stratified gender edifice construct the school atmosphere in which girls can be enrolled in record books without being in reality classes. In the classroom, the castes are still active. Children in Scheduled Castes, Scheduled Tribes and Other Backward Classes are also similarly discriminated against either implicitly or explicitly in confidence and learning. Cases of caste segregation in mid-day meals, unequal sitting, and biased attitude of teachers in various states have also been reported. Not only do such discriminatory

²⁸ Ekansh "Education Poverty: Understanding the Challenges and Solutions," available at: https://blog.goedu.in/education-poverty-challenges-and-solutions/?utm_source (last visited on 12.02.2026).

practices hinder academic progress, but they also strengthen psychological trauma and restrict the sense of belonging and involvement in children.²⁹

The disabled children face difficulties in mobility, communication and social integration. Numerous schools do not have resource teachers, available structure, or inclusive teaching. Although emphasis of the RTE Act is on inclusive education, the disparity between policy and practice is still broad. Learners with visual, hearing, and intellectual disabilities usually rely on specialised schools, which, however, most government schools do not offer. Parents with such children often testify to feeling marginalized by the mainstream education system because of the inadequacy of the infrastructural facilities as well as the stigma of the society.

The other socio-economic issue, which interferes with schooling, is migration. The seasonal migrants, such as the construction workers, agricultural workers or brick kiln workers tend to switch locations several times a year. They have their children facing frequent interruptions at school and thus learning continuity is a challenge. States have started seasonal hostels or traveling schools, but not all of them have equally taken the initiative. These barriers are important to understand so as to make policies that are ground-level real. The education policy should be aware of these overlapping socio-economic issues where access, retention and learning is not just assured in paper but being realised in the day to day life of children. It is only then that the RTE Act can have its transformative potential.

THE ROLE OF TECHNOLOGY: OPPORTUNITY AND INEQUALITY

Technology can transform education, but it may also increase the inequalities that exist. This duality was more pronounced than at any previous time due to the pandemic. Although the urban students shifted to online learning quite easily, millions of children in rural areas and low-income individuals were not included. In most sections of India, children used to share smartphones, intermittent mobile connections, or borrowed phones where possible. In the case of families having several school-going children, online classes were almost impossible to access. The outcome was to increase the learning gap that was disproportionately affecting the marginalised groups.³⁰

²⁹ Adv. Sandra Sachin, "Education, Poverty and Economic Development," available at: https://www.shekhar-law-office.com/articles/is-breaking-up-a-crime?utm_source (last visited on 11.01.2026).

³⁰ Sudeep A.M. & Akshay M., "A Study on the Impact of Poverty on Educational Access and Student Achievement in India," available at: https://tjjer.org/tjjer/papers/TIJERD001022.pdf?utm_source (last visited on 21.01.2026).

The RTE Act that has been prepared in 2009 fails to take the issue of technology-based education properly into consideration. Digital literacy and internet penetration were not the focus of the definition of quality education at the moment of its enactment and were low. Therefore, the Act does not specify the provision of digital infrastructure and schools and governments unprepared to follow technological-based models of learning are left. These restrictions became clear during the pandemic when most state governments rushed to implement online courses, video classes, or learning via mobile devices, yet lacked a legal framework that would allow imposing inconsistency and inequality in delivery.³¹

Digital divide is not just a device issue since it involves electricity, access to internet, digital literacy as well as an enabling learning atmosphere at home. The electricity supply is still problematic in most rural places, so children have no opportunity to charge their devices or use virtual classes. Low-speed networks are a hindrance to effective learning even in cases where connectivity is available. Also, digital literacy of parents especially in the marginalised communities is low and thus limits their capacity to mentor or assist the learning of their children. Gender norms led to denying girls access to digital devices in certain homes because girls were not encouraged to use mobile phones or because boys were traditionally given education.³²

In addition to infrastructure, the socio-emotional setting is also important in digital learning. A large number of the children do not have a place to study quietly, parental supervision and support at home. This atmosphere has direct impact on the concentration, motivation and learning outcomes. These underlying problems cannot be overlooked and turned into a privilege and not a right of digital education. Going forward, technology has to be incorporated at a meaningful level as opposed to a superficial one. Classroom teaching should not be substituted with digital tools. The hybrid models could add value to learning by enabling the students to watch video lessons, practice exercises, and interactives at their own pace. The integration, however, requires some careful planning rather than implementation at will.

Digital pedagogy training of teachers is essential. Lack of training on digital tools, instructional design, and online classroom management was one of the reasons why many teachers were struggling to cope with the shift to online platforms. Training programs should not be confined to simple device manipulation but aim to produce interactive learning programs that are student-

³¹ Leah Rodriguez "Understanding How Poverty is the Main Barrier to Education," available at: <https://www.globalcitizen.org/en/content/poverty-education-satistics-facts/> (last visited on 21/01/2026).

³² Vimala Ramachandran, *Public Education in India: Challenges and Prospects* (Routledge 2016).

centred. The access gap can be closed with the help of investments in infrastructure, including the digital labs in schools, community internet centres and public Wi-Fi zones. Panchayats and the local governments can actively participate by availing community facilities to learn digitally especially in remote areas.

Lastly, the mechanisms of monitoring within the RTE framework have to be modified to involve digital access and digital literacy as indicators of quality in the education. Technology can only democratize education through a holistic, equity-based approach, and help realize the vision of RTE Act in the modern-day world.

CONCLUSION: TOWARDS A MORE EFFECTIVE FUTURE FOR THE RTE ACT

The RTE Act remains one of India's most progressive social legislations, but its effectiveness is limited by structural, socio-economic, and administrative challenges. While the Act has succeeded in increasing enrolment, raising awareness, and initiating crucial conversations about educational equity, it has not fully achieved its core objective of ensuring meaningful, equitable, and high-quality education for every child. The gap between legislative intent and grassroots reality continues to be wide, particularly for children from marginalised socio-economic backgrounds, remote regions, and vulnerable social groups.

A reimagined future for the RTE Act must begin with acknowledging that access is only the first step learning, inclusivity, and empowerment must follow. The Act must evolve beyond the traditional focus on inputs and compliance to embrace a learning-outcome-centric approach. This requires systemic reforms that strengthen the pillars of education: teachers, infrastructure, pedagogy, governance, and community participation.

Teacher reforms must form the backbone of future efforts. Well-trained, motivated, and adequately supported teachers are essential for improving learning outcomes. This includes investing in continuous professional development, reducing non-academic burdens, filling teacher vacancies, and enhancing accountability through supportive, not punitive, mechanisms. Teachers should be empowered as leaders of change, not restricted to administrative checklist compliance.

Infrastructure development must also be viewed not merely as a construction exercise but as a commitment to dignity, safety, and an enabling environment. Functional toilets, electricity, digital access, libraries, playgrounds, and secure buildings are not privileges they are educational rights.

Strategic investments, transparent utilisation of funds, and decentralised monitoring can lead to sustained improvements.

The need to integrate technology responsibly and equitably is another crucial dimension. While digital tools hold immense potential, their benefits can only be realised when access is universal and digital literacy is widespread. The RTE Act must incorporate technology standards and digital-learning entitlements to ensure that no child is left behind in an increasingly digital world.

Community participation, too, must be strengthened. School Management Committees have tremendous potential to democratise governance, but they require capacity-building, awareness, and genuine decision-making power. When parents, teachers, and local communities collaborate, educational outcomes naturally improve.

Most importantly, inclusivity must be placed at the heart of educational reform. This means confronting gender disparities, caste discrimination, disability exclusion, and socio-economic barriers head-on. The next phase of the RTE must commit to intersectional justice, ensuring that every child, regardless of identity or background, has an equal chance to learn and succeed.

Education is not merely a legal right, it is a foundation for social justice, economic development, and democratic participation. Strengthening the RTE Act and addressing ground-level realities is essential for ensuring that every child in India not only attends school but truly learns, grows, and thrives. The promise of the RTE Act can be fulfilled only through sustained political will, informed policy reforms, community ownership, and a national commitment to placing children at the centre of all educational planning.

LEGITIMISING THE CONCEPT OF OPEN ADOPTION IN INDIA

- Prof. (Dr.) Achina Kundu* & Dr Baloy Bhattacharjee**

Abstract

Open adoption, where adoptive families maintain some level of contact with birth parents, is gaining recognition globally. In India, traditional closed adoption practices prevail, but there is a growing need to explore open adoption's potential benefits. Adoption laws in India, governed by the Hindu Adoptions and Maintenance Act, 1956, and the Juvenile Justice (Care and Protection of Children) Act, 2015, traditionally emphasise severing birth family ties upon adoption. However, the concept of "open adoption" is gaining traction internationally. Open adoption allows adopted children to maintain contact with their birth families, enabling them to understand their history, cultural roots, and access crucial medical information. Open adoption arrangements vary, with some countries like the US allowing post-adoption contact agreements between birth and adoptive parents. In the UK, "letterbox contact" facilitates limited communication. The England and Wales Court of Appeal recently emphasised the court's role in deciding post-adoption contact, disagreeing that adoption means complete severance of birth family ties. Advantages of open adoption include enhanced child identity, reduced feelings of loss, and access to medical history. In India, the Juvenile Justice Act allows adoption by relatives, potentially maintaining some birth family connection. Policymakers must balance the child's psychological stability with the potential benefits of open adoption.

Keywords: *Open Adoption, Birth Family, Adoptive Family, Child Welfare, Post-Adoption Contact, Letter box Contact.*

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INTRODUCTION

Adoption as an institution has been present since ancient times, involving the severance of a child from its birth or biological family and their placement into an adoptive family. Thus, adoption signifies the establishment of a legal relationship between individuals who are not biologically related as parent and child. While it is difficult to determine the first instance of adoption in India or ascertain its precise origins, references to adoption can be found in the Rigveda and other Vedic scriptures. During the Aryan period, the primary purpose of adoption was to ensure family fulfilment and the continuity of lineage, with a focus on the rights of adoptive parents. However, without delving into the evolution of adoption laws in India—a topic that merits separate discussion—it is noteworthy that the focus of adoption has shifted significantly over time. Today, the emphasis is on being child-centric, prioritising the ‘best interests’ and ‘welfare’ of the child rather than merely serving complete families.

In India, adoption is governed by two key legislations: the Hindu Adoptions and Maintenance Act, 1956, which applies specifically to Hindus, and the Juvenile Justice (Care and Protection of Children) Act, 2015, which facilitates the adoption of orphans, abandoned, and surrendered children, regardless of religious background. This paper, however, discusses the implications of severing relationships with a child’s birth family upon adoption and discourses on the prospects of maintaining contact between adopted children and their birth families. It explores whether such contact enhances the welfare and best interest policies established for the child. This discussion aims to ascertain the potential benefits and challenges of preserving these familial connections in the context of post-adoptive placements.

Key Objectives:

- To examines the implications of severing ties with a child's birth family upon adoption in India.
- To explores the prospects of maintaining contact between adopted children and their birth families.

REVIEW OF LITERATURE

Adoption in India - the Past, Present and the Future Trends written by Saras Bhaskar has rightly pointed out that Adoption in India has evolved significantly from traditional practices of

adopting a male child for performing last rites to a more progressive approach focusing on child welfare. The key points of the articles are as follows:

- Adoption has been practised in India for thousands of years, with references in Hindu epics like the Ramayana and the Mahabharata.
- Hindu Adoption and Maintenance Act (HAMA), 1956, governs adoption for Hindus, Buddhists, Jains, and Sikhs.
- Juvenile Justice (Care & Protection of Children) Act, 2000 allows secular adoption for all citizens.
- The Guardians and Wards Act, 1890, applies to Muslims, Christians, Parsis, and Jews for guardianship.

If we are talking about Trends and Challenges, the author has pointed out that:

- Domestic adoption has gained momentum with government regulation.
- Inter-country adoption was more prevalent due to lack of uniform laws and social stigma.
- Challenges include private adoptions, lack of data, and cultural preferences for male children.

Adoption laws in India are primarily governed by the Hindu Adoption and Maintenance Act (HAMA), 1956 for Hindus, Buddhists, Jains, and Sikhs. This act allows for the adoption of a child, giving the adopted child the same rights as a biological child.

Whereas the following key Points on Adoption in India are found in the article Adoption Laws in India: Challenging Existing Law” written by Niraj Meena

a) Hindu Adoption and Maintenance Act (HAMA), 1956:

- Applies to Hindus, Buddhists, Jains, and Sikhs.
- Allows for adoption of a child with rights similar to a biological child.
- Restrictions on adopting a child of a particular sex if parents already have a child of that sex.

b) Guardians and Wards Act, 1890:

- Applies to Muslims, Christians, Parsis, and Jews who can't adopt under their personal laws.
- Establishes a guardian-ward relationship rather than full adoption rights.

c) Juvenile Justice (Care & Protection of Children) Act, 2000:

- Applicable to all Indian citizens.
- Allows adoption of children, conferring parent-child status.

d) Adoption Across Religions in India:

- Hindus, Buddhists, Jains, and Sikhs follow HAMA for adoption.
- Muslims, Christians, Parsis, and Jews use the Guardians and Wards Act for guardianship, not full adoption.

In the article¹ “Bridging the Divide: Openness in Adoption and Post-adoption Psychosocial Adjustment among Birth and Adoptive Parents” written by J Fam Psychol & et.al., study examines the association between the degree of adoption openness and post-adoption adjustment among birth and adoptive parents. Using data from 323 matched pairs of birth mothers and adoptive parents, the study found that openness was significantly related to satisfaction with the adoption process among adoptive parents and birth mothers. Increased openness was positively associated with birth mothers' post-placement adjustment. Where it is found that:

- Openness was associated with greater satisfaction with the adoption process among adoptive parents and birth mothers.
- Increased openness was positively associated with birth mothers' post-placement adjustment.
- Birth fathers' report of openness was associated with their greater satisfaction with the adoption process and better post-adoption adjustment.

Relating to present trends, the article discusses the shift from closed to open adoptions and the debate surrounding the implications of openness on post-placement adjustment of birth and adoptive parents. Proponents of open adoption suggest benefits like increased security for

¹ J Fam Psychol, “Bridging the Divide: Openness in Adoption and Post-adoption Psychosocial Adjustment among Birth and Adoptive Parents” 22 (4) *Journal of Family Psychology* 529 (2008), available at: <https://doi.org/10.1037/a0012817> (last visited on: 24.11.2026).

adoptive parents and reduced grief for birth mothers. Opponents argue that openness might impede attachment and bonding between adoptive parents and the child.

Effects of Adoption

Section 12 of the Hindu Adoptions and Maintenance Act, 1956, emphatically states:

“An adopted child shall be deemed to be the child of his or her adoptive father or mother for all purposes with effect from the date of the adoption and from such date all the ties of the child in the family of his or her birth shall be deemed to be severed and replaced by those created by adoption in the adoptive family”²

Furthermore, Section 63 of the Juvenile Justice (Care and Protection of Children) Act, 2015 enunciates:

“A child in respect of whom an adoption order is issued..., shall become the child of the adoptive parents, and the adoptive parents shall become the parents of the child as if the child had been born to the adoptive parents, for all purposes, including intestacy, with effect from the date on which the adoption order takes effect, and on and from such date all ties of the child in the family of his or her birth shall be severed and replaced by those created by the adoption order in the adoptive family.”³

Severing and replacing the ties with the birth family can be easily understood – it means that the adopted child no longer bears any familial relationships with its birth family. Certain rights, especially in relation to succession and devolution of property, are retained. However, for the purposes of this paper, this topic does not merit a discussion.

Notwithstanding the above, is it plausible to enable the adopted child to remain in contact, where possible, with its birth family? The following section highlights the importance of this.

A GROWING RECEPTION TOWARDS OPEN ADOPTION

The Cambridge Dictionary defines ‘open adoption’ as “an arrangement by which children are legally adopted by people who are not their natural parents but still continue to communicate with their natural parents”.

² The Hindu Adoptions and Maintenance Act, 1956 (Act 78 of 1956), s.12

³ The Juvenile Justice (Care and Protection of Children) Act, 2015 (Act 2 of 2016), s.63

In broad terms, open adoption refers to an adopted child's birth family staying in touch with them. It covers a range of potential arrangements in which birth parents or other family members may stay in touch.

Although no legal definition of the topic is available, even in international law, several countries are increasingly acknowledging the concept. Although no country has formalised the idea into a legal framework, it is slowly gaining traction. For instance, in the US, it is common to see post-adoption contact agreements between birth parents and adoptive parents, which often formalise the obligations of the parties towards each other once the adoption has been finalised. The enforceability of such agreements differs from state to state, and most often, the child's best interest will be the overriding consideration.

In the United Kingdom, the concept of 'letterbox contact' has long been in practice, allowing adopted children to maintain limited contact with their birth families or significant family members through letters, carefully arranged visits, or sometimes newsletters from the adoptive parents to the birth family, which include updates on the child's growth and development.

The England and Wales' Court of Appeal's recent decision in *R & C (Adoption or Fostering)*⁴ is particularly remarkable in this context. The court considered whether there should be ongoing contact between the adopted child and its siblings and, in doing so, emphasised the court's ultimate responsibility, as opposed to that of the local authority or adoptive parents, in setting the tone for post-adoption contact. The court also disagreed with the trial judge's position that adoption meant the complete and irrevocable severance of all ties with the birth family.

ADVANTAGES OF OPEN ADOPTION

The primary advantage of open adoption is that it allows the adopted child to be aware of their history and gain a cultural awareness of their roots. It may also lessen the child's feelings of being lost and rejected, which is often the case, particularly with adolescent children.

Furthermore, remaining in touch with the birth family would help adopted children to have access to crucial medical history within the family, especially for hereditary diseases and illnesses.

In modern times, with the growing prevalence of social media, it may be nearly impossible to completely and irrevocably sever ties between birth families and adopted children when

⁴ (2024) EWCA Civ 1302

individuals can be reached with the click of a button. To keep up with these changes, it becomes essential to consider legitimising open adoption.

OPEN ADOPTION UNDER THE SCHEME OF THE JJ ACT

The evolution of the Juvenile Justice Act since its initial enactment in 1986 has changed the landscape of adoption in India by particularly enabling non-Hindus to adopt children who have been abandoned, surrendered or orphaned. This position has been reinforced by case laws of the Hon'ble Apex Court of India, where Muslims have been permitted to adopt under the JJ Act.⁵ However, this is not the only salient feature of the Act.

The most recent re-enactment of the JJ Act in 2015 also allows adoption by relatives of the child. This procedure may be particularly beneficial in cases of orphaned children who, if adopted by relatives, will retain some connection to their original family. The question of open adoption or maintaining contact with the birth family may not be as pertinent in the context of orphaned children, especially in relation to the birth parents, who would presumably be deceased. This brings us to the consideration of post-adoption contact in cases of abandoned and surrendered children. However, before discussing the same, it is essential to distinguish between the two.

Abandoned children are those whose parents or guardians have deserted them, and often, it may not be possible to trace the parent or guardian responsible for their abandonment. Surrendered children, in contrast, are those whose parents or guardians have given them up due to physical, emotional, or social factors beyond their control. In both cases, it remains to be determined whether the biological parents can be identified and if they wish for their identity and responsibilities towards the child to be assumed by an adoptive family while maintaining limited contact. The primary challenge in such situations would be, of course, the potential for creating instability and confusion in the child's life, particularly if the child is very young. The judiciary would certainly need to play a robust role in addressing legislative gaps should such a concept ever receive approval from Parliament. However, having discussed the advantages of open adoption, this is undoubtedly a consideration for law and policymakers.

Under the Hindu Adoptions and Maintenance Act 1956, the difficulty of tracing the birth parents or significant family member may not be as pertinent as adoption under Hindu Law is not limited to that of abandoned, surrendered and orphaned children. Regardless, the concerned

⁵ *M/s Shabnam Hashmi v. Union of India*, AIR 2014 SC 128. The Hon'ble Supreme Court, in this case, held that the JJ Act is an enabling legislation aimed at achieving the objectives of the Uniform Civil Code. Therefore, persons of any religion could adopt under this framework, subject to the applicable rules.

authorities would need to play a careful balancing act while maintaining the child's psychological stability, whilst giving effect to the advantages of open adoption.

JUDICIAL PRONOUNCEMENTS

In the following cases, the Supreme Court gave several judgments regarding the protection of the rights of a child, like in *Shabnam Hashmi v. Union of India*⁶, in *Sawan Ram v. Kalawanti*⁷, in *Srinivas Krishnarao Kango v. Narayan Devji Kango*⁸, in *Laxmi Kant Pandey v. Union of India*⁹, *Philips Alfred Marvin v. V. J. Gonsalves*¹⁰, *Rahasa Pandiani (Dead) by Lrs. v. Gokulananda Panda*¹¹, *A.S. Sailaja v. Principal, Kurnool Medical*¹².

CONCLUSION

In conclusion, the discourse on open adoption in India emphasises a significant shift towards prioritising the well-being and best interests of the child. As legal frameworks in other countries begin to recognise and formalise arrangements that allow for ongoing communication between adopted children and their birth families, there is a vital opportunity for India to explore similar practices. Embracing open adoption could not only enrich the adopted child's identity but also provide emotional support by preserving key familial connections.

Given the complexities and challenges of post-adoption relationships, policymakers and stakeholders must consider the diverse needs of children, adoptive parents, and birth families. By fostering a dialogue around open adoption, India can create a more inclusive, empathetic approach to adoption that aligns with contemporary understandings of family, attachment, and the child's best interests. Thus, the exploration of open adoption paves the way for a more holistic understanding of adoption practices in India, ultimately benefiting all parties involved.

⁶ AIR 2014 SC 1281

⁷ AIR 1967 SC 1761

⁸ AIR 1954 SC 379

⁹ AIR 1984 SC 469

¹⁰ AIR 1999 Ker 187

¹¹ AIR 1987 SC 962

¹² AIR 1986 AP 209

CIVIL NUCLEAR ENERGY AND INTERNATIONAL LAW: FRAMEWORKS, OBLIGATIONS, AND EMERGING CHALLENGES

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Abstract

The peaceful use of nuclear energy is one of the most consequential intersections between science, sovereignty, and international law. This article examines the normative architecture governing civil nuclear energy at the international level, tracing the evolution of legal obligations from the foundational Nuclear Non-Proliferation Treaty (NPT) of 1968 through the work of the International Atomic Energy Agency (IAEA) and beyond. It analyses the right to peaceful use of nuclear energy, the principle of non-proliferation, the role of safeguards agreements, liability regimes for nuclear damage, and the emerging legal questions surrounding nuclear security, radioactive waste management, and climate-driven nuclear expansion. The article also addresses the relevance of these frameworks to emerging nuclear states, with particular reference to India's unique legal position outside the NPT. It concludes by identifying areas requiring progressive development of international law to meet the challenges of the twenty-first century.

Keywords: *Nuclear Non-Proliferation Treaty; IAEA Safeguards; Peaceful Use of Nuclear Energy; Vienna Convention on Nuclear Liability; Nuclear Security; India-US Civil Nuclear Deal; International Nuclear Law*

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INTRODUCTION

Nuclear energy occupies a paradoxical space in the international legal order. Born from the weapons programme that devastated Hiroshima and Nagasaki in 1945, it was rapidly recast as a technology of development and peace through President Eisenhower's landmark 'Atoms for Peace' address before the United Nations General Assembly in December 1953. That address heralded the creation of the International Atomic Energy Agency (IAEA) in 1957 and laid the groundwork for a comprehensive international legal regime governing the civilian use of atomic power.

Today, nuclear energy supplies approximately ten per cent of global electricity generation and is increasingly positioned as a low-carbon option in the transition away from fossil fuels. As of 2024, thirty-two countries operate nuclear power plants, with a further twenty countries actively pursuing civil nuclear programmes. Against this backdrop, international law governing civil nuclear energy has grown from a sparse set of treaty commitments into a multi-layered regime encompassing non-proliferation obligations, safeguards, civil liability, physical protection, radioactive waste management, and emergency preparedness.

This article provides a comprehensive survey of the international legal framework applicable to civil nuclear energy, identifies its principal normative pillars, and examines tensions and lacunae that challenge its continued efficacy. Part II traces the foundational architecture. Part III analyses safeguards and verification. Part IV examines nuclear liability law. Part V addresses nuclear security. Part VI considers radioactive waste and decommissioning. Part VII discusses India's exceptional legal position. Part VIII identifies emerging challenges before concluding observations in Part IX.

THE FOUNDATIONAL LEGAL ARCHITECTURE

A. The Nuclear Non-Proliferation Treaty, 1968

The Treaty on the Non-Proliferation of Nuclear Weapons (NPT), which entered into force on 5 March 1970, constitutes the cornerstone of the international non-proliferation regime. Its architecture rests upon three mutually reinforcing pillars: non-proliferation, disarmament, and the inalienable right to the peaceful use of nuclear energy.

Article I obliges nuclear-weapon States (NWS) - defined in the Treaty as those that manufactured and exploded a nuclear device before 1 January 1967, namely the United States, Russia, the United Kingdom, France, and China - not to transfer nuclear weapons or control

over them to any recipient, nor to assist non-nuclear-weapon States (NNWS) in acquiring such weapons. Article II obliges NNWS not to receive, manufacture, or otherwise acquire nuclear weapons.¹

Crucially, Article IV preserves the inalienable right of all States Parties to develop research, production, and use of nuclear energy for peaceful purposes.² This provision has been the subject of sustained interpretive controversy: does the right to peaceful use extend to uranium enrichment and plutonium reprocessing - technologies that are inherently dual-use? The negotiating history and subsequent State practice suggest that Article IV encompasses the full fuel cycle, subject to safeguards obligations under Article III. However, the international community has increasingly sought to limit the spread of enrichment and reprocessing technology through supplier-group arrangements such as the Nuclear Suppliers Group (NSG).³

As at 2024, the NPT counts 191 States Parties, making it one of the most widely adhered-to arms control treaties in history. India, Pakistan, and Israel remain outside the Treaty, while North Korea purported to withdraw in 2003, though the legal validity of that withdrawal is contested.

B. The International Atomic Energy Agency

Established by the IAEA Statute in 1956, the Agency is mandated to accelerate and enlarge the contribution of atomic energy to peace, health, and prosperity throughout the world, while ensuring that its assistance is not used to further any military purpose.⁴ The IAEA performs three core functions: safety standards-setting, technical cooperation, and verification through safeguards.

The Agency's legal authority derives from a network of agreements including: (i) safeguards agreements with Member States under NPT Article III; (ii) the Additional Protocol (INFCIRC/540) which enhances inspection rights; (iii) project and technical cooperation agreements; and (iv) bilateral nuclear cooperation agreements.⁵ The IAEA Board of Governors and General Conference constitute its principal decision-making organs, while the Director General serves as the chief administrative officer.

¹ Treaty on the Non-Proliferation of Nuclear Weapons, 1 July 1968, 729 UNTS 161, Arts. I–II.

² *Ibid*, Art. IV.

³ Nuclear Suppliers Group, 'Guidelines for Nuclear Transfers' (INFCIRC/254/Rev.14/Part 1, IAEA, Vienna, 2022).

⁴ IAEA Statute, 26 October 1956, 276 UNTS 3, Art. II.

⁵ IAEA Statute (n 3) Art. III(A)(5); C. Stoiber et al., Handbook on Nuclear Law (IAEA, Vienna, 2003) 15-22.

The IAEA issues legally non-binding but practically authoritative Safety Standards in the form of Safety Fundamentals, Safety Requirements, and Safety Guides. These standards, developed through expert consensus, are incorporated into domestic law by most operating States and inform bilateral nuclear cooperation agreements globally.

Table 1: Key International Legal Instruments in Civil Nuclear Energy

Instrument	In Force	Principal Subject Matter	Parties
Treaty on the Non-Proliferation of Nuclear Weapons (NPT)	1970	Non-proliferation, disarmament, peaceful use	191
IAEA Statute	1957	Establishment of IAEA; safeguards mandate	176
Vienna Convention on Civil Liability for Nuclear Damage	1977/2003	Third-party nuclear liability	39
Paris Convention on Nuclear Third Party Liability	1968	Western European liability regime	16
Convention on Nuclear Safety	1994	Safety obligations for land-based reactors	92
Joint Convention on Safety of Spent Fuel	2001	Waste management obligations	87
Convention on Physical Protection (CPPNM/2005 Amend.)	2016	Physical security of nuclear material	163
CSA (INFCIRC/153)	Ongoing	State-level IAEA safeguards	182

Instrument	In Force	Principal Subject Matter	Parties
Additional Protocol (INFCIRC/540)	Ongoing	Enhanced verification rights	140+
Convention on Supplementary Compensation (CSC)	2015	Supplementary nuclear liability fund	12

C. Nuclear-Weapon-Free Zone Treaties

Regional Nuclear-Weapon-Free Zone (NWFZ) treaties complement the global NPT regime by prohibiting the stationing, testing, and use of nuclear weapons within defined geographic areas while explicitly affirming the right to peaceful nuclear energy. The Treaty of Tlatelolco (1968, Latin America and the Caribbean), the Treaty of Rarotonga (1986, South Pacific), the Treaty of Bangkok (1997, Southeast Asia), the Treaty of Pelindaba (2009, Africa), and the Treaty of Semipalatinsk (2009, Central Asia) collectively establish substantial NWFZ coverage across the Global South. These instruments create positive obligations for civil nuclear cooperation and negative security assurances from NWS Parties.

SAFEGUARDS AND VERIFICATION

A. The Comprehensive Safeguards Agreement

Article III of the NPT obliges each NNWS Party to conclude a Comprehensive Safeguards Agreement (CSA) with the IAEA. The model text for CSAs - INFCIRC/153 - requires States to declare all nuclear material subject to the Treaty and accept IAEA verification through on-site inspections, material accountancy, and containment and surveillance measures.⁶ The IAEA verifies the non-diversion of declared nuclear material to non-peaceful purposes.⁷

However, the CSA system was exposed as inadequate by Iraq's clandestine weapons programme discovered after the 1991 Gulf War. Iraq had declared its safeguarded facilities accurately but pursued covert enrichment outside the declared programme. The IAEA lacked authority under

⁶ IAEA, 'The Structure and Content of Agreements Between the Agency and States Required in Connection with the Treaty on the Non-Proliferation of Nuclear Weapons' (INFCIRC/153 (Corrected)), IAEA, Vienna, 1972) para. 1.

⁷ Ibid, paras. 2-3.

the CSA to investigate undeclared activities. This vulnerability prompted negotiation of the Additional Protocol.

B. The Additional Protocol

The Model Additional Protocol (INFCIRC/540), approved by the IAEA Board of Governors in 1997, significantly expands the IAEA's verification authority. Key innovations include: (i) expanded declarations covering the entire nuclear fuel cycle, including uranium mining and fuel fabrication; (ii) short-notice complementary access to any location a State declares or the IAEA deems relevant; (iii) environmental sampling at and beyond declared sites; and (iv) collection of information on nuclear fuel cycle research and development.⁸ The Additional Protocol has become the de facto standard of verification adequacy, with many nuclear supplier States making its conclusion a condition for nuclear cooperation. As of 2024, over 140 States have the Additional Protocol in force.⁹

C. State-Level Concept and Integrated Safeguards

The IAEA has progressively developed a State-Level Concept (SLC) which evaluates the absence of undeclared nuclear material and activities in a State as a whole, rather than limiting analysis to declared facilities.¹⁰ States for which the IAEA has drawn a credible assurance of absence of undeclared material - the 'broader conclusion' - may be eligible for Integrated Safeguards (IS), which rationalises inspection frequency based on verified risk assessment. IS represents a sophisticated evolution of the verification system, moving from accountancy-based to intelligence-informed assurance.¹¹

INTERNATIONAL NUCLEAR LIABILITY LAW

A. The Vienna-Paris Framework

The international law of nuclear liability is governed by two principal instruments: the Vienna Convention on Civil Liability for Nuclear Damage (1963, as revised in 1997) and the Paris Convention on Third Party Liability in the Field of Nuclear Energy (1960, as revised). A Joint

⁸ IAEA, 'Model Protocol Additional to the Agreement(s) between State(s) and the International Atomic Energy Agency for the Application of Safeguards' (INFCIRC/540 (Corrected), IAEA, Vienna, 1997) Art. 5.

⁹ Ibid, Arts. 4–5.

¹⁰ IAEA, 'Safeguards Implementation Report for 2022' (GOV/2023/23, IAEA, Vienna, 2023) 7.

¹¹ IAEA, 'The Evolution of IAEA Safeguards' (International Verification Series No. 2, IAEA, Vienna, 1998) 18-24.

Protocol concluded in 1988 extends the benefits of each convention to nationals of the other, creating a degree of inter-system coherence.¹²¹³

Both instruments rest upon common principles: (i) absolute liability of the nuclear operator regardless of fault; (ii) exclusive liability, channelled to the licensed operator; (iii) limitation of liability in amount and time; and (iv) exclusive jurisdiction of the courts of the installation State.¹⁴ The 1997 Vienna Protocol substantially revised the original text, extending the geographical scope of compensable damage to include the exclusive economic zone, adding environmental damage and preventive measures as compensable heads, and raising the liability ceiling.

B. The Convention on Supplementary Compensation

The Convention on Supplementary Compensation for Nuclear Damage (CSC), adopted by the IAEA in 1997 and in force since 2015, creates a two-tier compensation system applicable regardless of whether the installation State is a party to the Vienna or Paris conventions. The first tier requires States to ensure a minimum national compensation fund. The second tier is a collectively-funded international pool drawn upon when first-tier funds are exhausted. The United States, Japan, India, Romania, and Morocco are among the Parties. The CSC is notable for its compatibility with national liability legislation that does not necessarily conform to the channelling model, making it more accessible to States with divergent domestic regimes.¹⁵

C. The Indian Civil Liability for Nuclear Damage Act, 2010

India's Civil Liability for Nuclear Damage Act, 2010 (CLNDA) generated significant international controversy upon its enactment. Section 17(b) of the CLNDA provides the operator a right of recourse against a supplier where the nuclear incident has resulted from a latent or patent defect in the supplied equipment or material,¹⁶ and Section 46 preserves the application of any other law in force.¹⁷ These provisions were widely criticised by foreign nuclear suppliers as incompatible with the exclusive liability and channelling principles of the CSC, to

¹² Vienna Convention on Civil Liability for Nuclear Damage, 21 May 1963, 1063 UNTS 265, as amended by the Protocol of 12 September 1997, Arts. II, IV.

¹³ Joint Protocol Relating to the Application of the Vienna Convention and the Paris Convention, 21 September 1988, 1672 UNTS 302.

¹⁴ Ibid, Arts. V, XI; Paris Convention on Third Party Liability in the Field of Nuclear Energy, 29 July 1960, 956 UNTS 251, as revised, Art. 3.

¹⁵ Convention on Supplementary Compensation for Nuclear Damage, 12 September 1997, IAEA Doc. INFCIRC/567, Arts. III–IV.

¹⁶ Civil Liability for Nuclear Damage Act, 2010 (India), s. 17(b).

¹⁷ Ibid, s. 46.

which India is a Party.¹⁸ The controversy has impeded implementation of the India-US Civil Nuclear Agreement and dampened commercial interest from American and French vendors. The Indian government has attempted to address supplier concerns through administrative arrangements, though their legal adequacy remains disputed.

Table 2: Comparative Overview of Nuclear Liability Regimes

Feature	Vienna Conv. (1997)	Paris Conv. (2004)	CSC (1997)	India CLNDA (2010)
Liability basis	Strict/Absolute	Strict/Absolute	Strict/Absolute	No-fault
Channelling	To operator	To operator	To operator	To operator (with recourse)
Liability limit	SDR 300 million	EUR 700 million	SDR 300 million (min.)	INR 1500 crore
Environmental damage	Covered	Covered	Covered	Covered
Supplier recourse	Not provided	Not provided	Not provided	Permitted (s. 17)
Geographic scope	EEZ included	EEZ included	EEZ included	National territory

NUCLEAR SECURITY LAW

A. The Convention on Physical Protection of Nuclear Material

¹⁸ P. Sands & P. Galizzi, 'The 1997 Vienna Convention on Civil Liability for Nuclear Damage' (1999) 48 International & Comparative Law Quarterly 475, 488.

The Convention on the Physical Protection of Nuclear Material (CPPNM), originally concluded in 1980 and significantly amended in 2005 (the Amendment entering into force in 2016), establishes binding obligations on States to protect nuclear material in international transport and, as amended, domestic use, storage, and transport.¹⁹ The 2005 Amendment was a direct response to the post-9/11 recognition that terrorist acquisition of nuclear or radiological material posed an existential threat. The Amendment extends CPPNM coverage to domestic nuclear facilities, requires States to enact legislation criminalising the sabotage of nuclear facilities, and establishes an obligation to share information relevant to the physical protection of nuclear material.

B. UN Security Council Resolution 1540 (2004)

Adopted under Chapter VII of the UN Charter, Security Council Resolution 1540 (2004) imposes legally binding obligations on all UN Member States to adopt and enforce effective measures against the proliferation of weapons of mass destruction and their means of delivery, with particular emphasis on preventing non-State actors from acquiring nuclear, biological, or chemical weapons.²⁰ Resolution 1540 requires States to enact domestic legislation criminalising the manufacture, acquisition, and transfer of such weapons and to adopt border controls and export regulations.²¹ The 1540 Committee monitors implementation.

C. Nuclear Security Summits and Soft Law

The Nuclear Security Summit (NSS) process, initiated by President Obama in 2010 and continuing through four summits until 2016, produced a substantial body of politically binding commitments - communiqués, national action plans, and gift baskets - that have shaped State practice significantly even without formal treaty character. The NSS process resulted in the removal of highly enriched uranium (HEU) from multiple States, accelerated ratification of the CPPNM Amendment, and strengthened bilateral nuclear security cooperation.²² The IAEA's Nuclear Security Series of guidance documents serves as the primary technical reference for States implementing their nuclear security obligations.

¹⁹ Convention on the Physical Protection of Nuclear Material, 3 March 1980, 1456 UNTS 125, as amended 2005, Art. 2A.

²⁰ UN Security Council Resolution 1540, 28 April 2004, S/RES/1540 (2004) para. 2.

²¹ Ibid, para. 3.

²² IAEA, 'Nuclear Security Series No. 13: Nuclear Security Recommendations on Physical Protection of Nuclear Material and Nuclear Facilities (INFCIRC/225/Revision 5)' (NSS No. 13, IAEA, Vienna, 2011).

RADIOACTIVE WASTE MANAGEMENT AND DECOMMISSIONING

A. The Joint Convention

The Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management, which entered into force in 2001, is the first international legal instrument to address the safety of civilian radioactive waste and spent fuel on a global scale.²³ Unique in its structure, the Joint Convention operates through a peer review mechanism: Contracting Parties submit national reports on the measures taken to implement their obligations, which are then reviewed by other Contracting Parties at triennial review meetings. This non-adversarial peer review model has proven effective in promoting continuous improvement.

B. Prohibition of Ocean Dumping

The 1972 London Convention and its 1996 Protocol prohibit the dumping of radioactive waste at sea.²⁴ The 1993 amendment to the London Convention established a moratorium on sea disposal of low-level radioactive waste, effectively extending the prohibition to all radioactive material. The evolution from permitting to prohibiting marine disposal reflects both advances in waste management technology and a precautionary approach to marine environmental protection. Under UNCLOS Article 194, States are also required to prevent, reduce, and control pollution of the marine environment from all sources, including land-based radioactive contamination.²⁵

INDIA'S LEGAL POSITION IN THE INTERNATIONAL NUCLEAR ORDER

A. India Outside the NPT

India occupies an anomalous position in international nuclear law. Having conducted its first nuclear test in 1974 (Pokharan-I) and declared itself a nuclear-weapon State following the 1998 Pokharan-II tests, India has never been party to the NPT. As an NPT non-party, India is not subject to IAEA comprehensive safeguards requirements under Article III. However, India's civilian nuclear facilities are subject to facility-specific safeguards under a 2009 India-IAEA

²³ Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management, 5 September 1997, 2153 UNTS 357, Art. 30.

²⁴ Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter, 29 December 1972, 1046 UNTS 120; 1996 Protocol thereto, Art. 4(1).

²⁵ UN Convention on the Law of the Sea, 10 December 1982, 1833 UNTS 397, Art. 194(3)(a).

Safeguards Agreement concluded pursuant to the India-specific exception granted by the Nuclear Suppliers Group in 2008.²⁶

B. The India-US Civil Nuclear Agreement (123 Agreement), 2008

The Agreement for Cooperation Between the Government of India and the Government of the United States of America Concerning Peaceful Uses of Nuclear Energy (123 Agreement), signed on 10 October 2008, represented a watershed in India's integration into the civil nuclear order. The Agreement was made possible by the Hyde Act (2006) in the United States,²⁷ which granted a waiver from standard US nuclear cooperation requirements, and by the NSG's grant of a State-specific exemption from its full-scope safeguards requirement.

Under the 123 Agreement, the United States and India agreed to cooperate in civil nuclear energy including nuclear fuel supply, reactor technology transfer, and reprocessing of US-obligated nuclear material under IAEA safeguards in a jointly agreed facility.²⁸ The Agreement also contains a right of return provision obliging India to return US-obligated nuclear material and equipment in the event of a nuclear test.²⁹ The Agreement has not been fully operationalised, primarily due to disagreements over the supplier liability provisions of India's CLNDA, 2010.³⁰

C. India as a Responsible Nuclear State

India has increasingly engaged with the international nuclear security architecture. India joined the International Convention for the Suppression of Acts of Nuclear Terrorism (2005),³¹ ratified the 2005 CPPNM Amendment (2017), and concluded the CSC (2016). India participates actively in IAEA technical cooperation programmes and contributes to the IAEA's nuclear security fund. India's nuclear doctrine of credible minimum deterrence and no-first-use, while not embedded in any international legal instrument, has been consistently articulated as a responsible nuclear posture. India's application for full membership of the NSG remains pending, contested primarily by China.

²⁶ IAEA, 'Agreement between the Government of India and the International Atomic Energy Agency for the Application of Safeguards to Civilian Nuclear Facilities' (INFCIRC/754, IAEA, Vienna, 2009).

²⁷ Henry J. Hyde United States-India Peaceful Atomic Energy Cooperation Act, Pub. L. No. 109-401, 120 Stat. 2726 (2006).

²⁸ Agreement for Cooperation Between the Government of India and the Government of the United States of America Concerning Peaceful Uses of Nuclear Energy, 10 October 2008 (123 Agreement), Art. 6.

²⁹ Ibid, Art. 14(4) (right of return clause).

³⁰ S.K. Singh, 'India's Civil Nuclear Programme: Law, Policy and Paradox' (2012) 52 Indian Journal of International Law 1, 14.

³¹ Convention for the Suppression of Acts of Nuclear Terrorism, 13 April 2005, 2445 UNTS 89.

D. India's Three-Stage Nuclear Programme: Legal Dimensions of the Fast Breeder Reactor

A defining feature of India's civil nuclear strategy is the three-stage nuclear programme, conceived by Dr. Homi J. Bhabha, which seeks to achieve long-term energy self-sufficiency through sequential exploitation of uranium and thorium resources. Stage I relies on Pressurised Heavy Water Reactors (PHWRs); Stage II deploys Fast Breeder Reactors (FBRs) using plutonium derived from Stage I operations; and Stage III envisages advanced thorium-based reactors. The legal significance of this programme lies in its intimate relationship with nuclear material production, reprocessing, and safeguards - all of which engage India's obligations under its IAEA Safeguards Agreement and the broader international non-proliferation architecture.

India's Prototype Fast Breeder Reactor (PFBR) at Kalpakkam, Tamil Nadu, achieved criticality on 6 April 2026.³² From an international law perspective, this milestone is significant because the PFBR is designated as a civilian facility under India's Separation Plan - the schedule of civilian and military nuclear facilities communicated to the IAEA pursuant to the 2008 NSG exemption and the 2009 India-IAEA Safeguards Agreement.³³ As a civilian facility, the PFBR is expected to be brought under IAEA facility-specific safeguards once it attains commercial operation, creating a direct legal obligation for India to declare the facility, accept Agency inspections, and maintain nuclear material accountancy.

The reprocessing of spent fuel from PHWRs to extract plutonium for the FBR programme is a legally sensitive activity. Reprocessing is a dual-use technology - it can produce plutonium for civilian reactor fuel or, with modification, weapons-grade material. Under the India-IAEA Safeguards Agreement, reprocessing of safeguarded nuclear material is subject to IAEA approval and verification. India's right to reprocess US-origin nuclear material is also addressed in the 123 Agreement, which provides for consultations on a dedicated reprocessing facility under IAEA safeguards.³⁴ The operationalisation of the PFBR will generate increased plutonium throughput and correspondingly heightened safeguards obligations.

The governance of India's nuclear programme also raises questions of legal accountability. The Department of Atomic Energy (DAE) reports directly to the Prime Minister's Office, and the nuclear sector operates with limited parliamentary oversight and judicial scrutiny. The Atomic

³² Vajiram & Ravi Mains Team, 'FBR in India: How FBR Strengthens India's Nuclear Energy Strategy', Vajiram & Ravi Current Affairs (9 April 2026) <<https://vajiramandravi.com/current-affairs/fbr-strengthens-indias-nuclear-energy-strategy/>> accessed 10 April 2026.

³³ IAEA, 'India's Separation Plan' (INFCIRC/731, IAEA, Vienna, 2008); INFCIRC/754 (n 28).

³⁴ 123 Agreement (n 30) Art. 6(iii); INFCIRC/754 (n 28) Art. 14.

Energy Act, 1962 concentrates regulatory and operational functions within the same institutional framework, an arrangement that has been criticised as falling short of the IAEA's Safety Requirements for independent nuclear regulatory authority.³⁵ The Convention on Nuclear Safety, to which India is not a Party, specifically requires the establishment of a regulatory body independent from entities promoting nuclear energy - a standard that India's present governance structure does not fully satisfy. Operationalisation of the PFBR will renew calls for legislative reform to separate regulatory from promotional functions in India's nuclear sector.

EMERGING CHALLENGES FOR INTERNATIONAL NUCLEAR LAW

A. Nuclear Energy and Climate Change

The Paris Agreement's ambition to limit global temperature increase to 1.5°C above pre-industrial levels has renewed interest in nuclear energy as a low-carbon electricity source.³⁶ The International Energy Agency and the IPCC have identified nuclear as a component of a decarbonised energy mix.³⁷ This nuclear renaissance raises important questions for international law: Does climate necessity alter the interpretation of Article IV of the NPT? Can climate finance instruments under the UN Framework Convention on Climate Change legitimately support civil nuclear programmes? How does the treaty regime adapt to novel reactor technologies, including small modular reactors (SMRs) and advanced reactor concepts that may require different safeguards approaches?

B. Small Modular Reactors and the Legal Framework

Small modular reactors (SMRs), defined as reactors with generating capacity below 300 MWe, are designed for factory fabrication and modular deployment.³⁸ Their characteristics - smaller footprint, lower upfront capital, potential for remote siting - raise novel regulatory and safeguards questions. Can IAEA safeguards designed for large light-water reactors be efficiently applied to hundreds of small reactors potentially deployed across developing countries? Who bears liability when a factory-fabricated reactor module supplied by a foreign vendor causes

³⁵ Atomic Energy Act, 1962 (India), s. 3; Convention on Nuclear Safety, 17 June 1994, 1963 UNTS 293, Art. 8(2).

³⁶ Paris Agreement under the United Nations Framework Convention on Climate Change, 12 December 2015, UN Doc. FCCC/CP/2015/L.9/Rev.1, Art. 2(1)(a).

³⁷ IEA, 'Net Zero by 2050: A Roadmap for the Global Energy Sector' (IEA, Paris, 2021) 103.

³⁸ IAEA, 'Advances in Small Modular Reactor Technology Developments' (IAEA, Vienna, 2022) 5.

nuclear damage in a host State? How do export control regimes apply to modular reactor components? These questions require proactive legal development.³⁹

C. Cybersecurity and Nuclear Facilities

The Stuxnet incident in 2010 demonstrated that industrial control systems at nuclear enrichment facilities could be compromised by sophisticated cyber weapons, exposing the intersection of cyberlaw and nuclear security. There is no specific international treaty addressing cyber attacks on nuclear facilities. The applicability of existing international humanitarian law principles - distinction, proportionality, precaution - to cyber operations against nuclear installations remains debated.⁴⁰ The Tallinn Manual 2.0 on the International Law Applicable to Cyber Operations addresses some of these questions in the context of the law of armed conflict but does not resolve peacetime obligations.⁴¹ The IAEA has issued non-binding Computer Security guidance (NSS No. 17-T) but the legal framework remains significantly underdeveloped.

D. Nuclear Disarmament and the Treaty on the Prohibition of Nuclear Weapons

The Treaty on the Prohibition of Nuclear Weapons (TPNW), adopted by 122 States in 2017 and in force since January 2021, prohibits the development, testing, production, stockpiling, use, and threat of use of nuclear weapons. Article 1(e) thereof prohibits assistance to any prohibited activity,⁴² raising questions about whether TPNW Parties may participate in certain forms of nuclear fuel cycle cooperation with NWS. All five NPT-recognised NWS, India, Pakistan, and Israel boycotted the TPNW negotiations.⁴³ The TPNW-NPT relationship, and the TPNW's compatibility with existing civil nuclear cooperation frameworks, constitutes a significant legal and political tension in the contemporary nuclear order.

E. Fast Breeder Reactors, Thorium Cycles, and Safeguards Adaptation

The advancement of fast breeder reactor technology and the prospective deployment of thorium-based fuel cycles present two of the most consequential legal challenges for international safeguards law in the coming decades. The standard IAEA Comprehensive

³⁹ M. Hibbs, 'The Future of Nuclear Power in East Asia' (Carnegie Endowment for International Peace, Washington DC, 2011) 44.

⁴⁰ T. Rid, 'Cyber War Will Not Take Place' (2012) 35 Journal of Strategic Studies 5, 18; IAEA, 'Computer Security for Nuclear Security' (NSS No. 17-T, IAEA, Vienna, 2021).

⁴¹ M.N. Schmitt (ed.), Tallinn Manual 2.0 on the International Law Applicable to Cyber Operations (Cambridge University Press, Cambridge, 2017) Rule 80.

⁴² Treaty on the Prohibition of Nuclear Weapons, 7 July 2017, UN Doc. A/CONF.229/2017/8, Art. 1(e).

⁴³ D.H. Joyner, 'The Legal Meaning and Implications of Article VI of the Non-Proliferation Treaty' (2011) 37 Yale Journal of International Law 259, 274.

Safeguards Agreement model (INFCIRC/153) and the verification techniques developed thereunder were designed primarily for light-water and heavy-water reactor technology. Fast breeder reactors involve continuous conversion of fertile material to fissile plutonium in the reactor blanket, a process that generates nuclear material accounting challenges distinct from those arising in conventional reactor operations.⁴⁴ The IAEA has acknowledged the need to develop safeguards approaches adapted to fast reactors, but no standardised legally operative methodology has yet been adopted.⁴⁵ This gap between legal obligation - the duty of non-diversion under Article III NPT - and technical verification capacity is a structural vulnerability in the present safeguards system.

India's PFBR at Kalpakkam, as it advances toward commercial operation under IAEA facility-specific safeguards, will be an early test case for the adequacy of existing verification law applied to an operating fast breeder reactor. The IAEA will need to negotiate a safeguards approach document with India specifying the inspection frequency, material balance areas, key measurement points, and accountancy procedures appropriate for the PFBR's fuel cycle. The legal principle of effective safeguards - requiring that verification measures provide timely detection of diversion of a significant quantity of nuclear material - must be operationalised through technically credible means for the fast reactor context.⁴⁶

The thorium fuel cycle raises a distinct legal gap. Thorium-232, as a fertile but non-fissile material, is currently not subject to IAEA safeguards obligations under INFCIRC/153, which applies to 'source material' (uranium and thorium) only when it is being processed for use in a nuclear reactor or after it has been irradiated.⁴⁷ Once irradiated, thorium-232 converts to uranium-233 - a fissile material with weapons-applicable characteristics - which does attract safeguards. However, the legal framework governing the accountancy, inspection, and verification of uranium-233 production in thorium reactors remains underdeveloped.⁴⁸ As India moves toward Stage III of its three-stage programme, the international community will face a pressing obligation to develop progressive law in this area, whether through amendment of safeguards agreements, new IAEA Board of Governors decisions, or a dedicated international instrument.

⁴⁴ INFCIRC/153 (n 8) paras. 74-78 (material balance areas and key measurement points).

⁴⁵ IAEA, 'Safeguards Approaches for Facilities: Fast Reactors' (STR-375, IAEA, Vienna, 2013) 3-9.

⁴⁶ Ibid, 12; INFCIRC/153 (n 8) para. 28 (timely detection standard: one significant quantity within conversion time).

⁴⁷ INFCIRC/153 (n 8) para. 34(c) (definition of source material including thorium).

⁴⁸ IAEA, 'Thorium Fuel Cycle - Potential Benefits and Challenges' (TECDOC-1450, IAEA, Vienna, 2005) 99-105.

CONCLUSION

The international legal framework governing civil nuclear energy has evolved substantially over seven decades, from the bare bones of the IAEA Statute and NPT to a dense web of treaty obligations, IAEA standards, export control guidelines, and soft-law commitments. This framework has demonstrably contributed to preventing the spread of nuclear weapons, promoting safety in nuclear operations, and establishing minimum standards of nuclear security globally.

Yet the framework faces mounting challenges. The fragmentation of the liability regime, exemplified by India's CLNDA departing from the standard channelling model, undermines the coherence of nuclear commerce. The non-universality of the NPT, the stalled disarmament pillar, and the absence of NWS from the TPNW create structural tensions. Emerging technologies - SMRs, advanced reactors, digital instrumentation - test the adaptability of verification and safety instruments developed for mid-twentieth-century reactor designs. Cyber vulnerabilities impose new threat vectors that existing legal instruments were not designed to address.

Progressive development of international nuclear law must address these lacunae through a combination of treaty amendment, new instruments where necessary, authoritative interpretation of existing obligations, and continued elaboration of IAEA standards and guidance. The universalisation of the Additional Protocol, the harmonisation of liability regimes through broader adherence to the CSC, and the negotiation of safeguards approaches adapted to SMR deployment deserve priority attention.

For emerging and re-emerging nuclear energy States, including India, the path forward lies in constructive engagement with the international nuclear legal order - not through seeking exception from its requirements but through contributing to its evolution. The inalienable right to peaceful nuclear energy, recognised in Article IV of the NPT and reflected in the foundational instruments surveyed here, carries with it an equal obligation of responsibility: to the international community, to future generations, and to the integrity of the non-proliferation regime upon which global nuclear order rests.

FROM PHISHING TO DEEPFAKES: EVALUATING INDIA'S PREPAREDNESS AGAINST EMERGING CYBER THREATS

- Anjali*, Satwik Bhardwaj**, & Dr Dhawal Shankar Srivastava***

Abstract

India has always been known for its rapid growth in every aspect, especially technology. India has witnessed a huge progress in technology. But as it continues to grow, a major concern of cybersecurity has also emerged. Earlier, there used to be basic phishing attacks, but now highly sophisticated attacks are taking place, powered by new technologies in India, which are difficult to detect. This paper critically explores the spectrum of these emerging cyber threats and critically evaluates whether India's legal and institutional framework is ready to tackle these threats. It examines India's current legislation on cybersecurity including Information Technology Act, 2000, Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021, Digital Personal Data Protection Act, 2023, Bharatiya Nyaya Sanhita, 2023, CERT In 2013 Rules, 2022 Directions and newly introduced IT (Intermediary Guidelines & Digital Media Ethics Code) Amendment Rules, 2026 along with efforts made by institutions like CERT-In, NCIIPC, and NCCC. The paper also highlights these cyberattacks through a combination of doctrinal analysis, which includes case studies and global comparative analysis. The paper argues that although IT Amendment Rules, 2026 aimed at modernizing India's cybersecurity framework in response to deepfakes, the approach continues to remain reactive and insufficient to address the growing risks posed by deepfakes. The paper also highlights that in order to effectively deal with modern cyber threats, prioritize cyber education and public awareness on these areas. This paper also assesses the India's position in Cybersecurity Ecosystem, offering a comparative analysis with leading international cybersecurity models. At the end of the paper, it has been suggested that India must take a shift from a reactive stance to a proactive cybersecurity governance model. This approach will not only protect the digital infrastructure of India but also preserve the integrity of its democratic institutions and fundamental rights of its citizens in an ever- changing cyber threat environment.

Keywords: *Artificial Intelligence, Cybersecurity, Deepfakes, IT Act, Phishing.*

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INTRODUCTION

A massive rise in rapid digitisation has been seen in the 21st century, which has entirely changed the digital ecosystem of India. India today has a digital population of nearly 850 million, which makes it one of the largest digital ecosystems in the world. This digital surge has been driven by initiatives like Aadhaar Unified Payments Interface (UPI), Digital India and e-governance. But this digital revolution comes with a drawback: the risk of increasing cyberthreats.

A Cyber threat is any potential danger that causes an unlawful activity in computer systems or computing devices, such as smartphones, tablets or networks, with the intention of stealing, disrupting or gaining unauthorised access, such as hacking, phishing or deepfakes. It is often committed by people with a destructive and criminal mindset, either for revenge or monetary gain. Cyber threats can happen in different forms, from using traditional ways such as hacking, phishing, to sophisticated ones such as deepfakes powered by artificial intelligence, which are harder to detect.

Phishing is the most common and traditional form of attacking people digitally. The commonly stolen data are login credentials and credit card numbers. The goal is to send an email or SMS that seems like something the victim needs or wants and induce him/her to click a link. The moment the victim clicks on the link, their personal data gets stolen from them.

Deepfakes are increasingly regarded as modern high-risk cyberweapons, producing AI-generated synthetic videos or audio that impersonate real people to spread misinformation or manipulate public opinion. They seem convincingly real but are actually fake. They blur the line between reality and fiction. Imagine receiving a video message from your CEO asking for an urgent wire transfer. You see their face. You hear their voice. Would you question it? That is the dangerous power of deepfakes; they can trick and manipulate people.

The shift from phishing to deepfakes highlights the increasing complexity of technologies used in cyberattacks, such as machine learning, artificial intelligence and big data analytics. The implications of cyber-attacks can be financial loss, privacy invasion, political manipulation, or even national security threats, which means cybersecurity is no longer a technical issue but a matter of democratic stability and social trust. In India, although there are laws like Information Technology Act, 2000, Digital Personal Data Protection Act, 2023, Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021, Bharatiya Nyaya Sanhita, 2023, the CERT In 2013 Rules and 2022 Directions which provides safeguards and institutions like CERT-In, NCIIPC, and NCCC, but these laws have no explicit legal provisions against

deepfakes. However, the newly introduced IT Amendment Rules, 2026, explicitly address deepfakes and focus on regulating AI-generated content, but continue to remain reactive and insufficient to address the growing risks posed by deepfakes.

OBJECTIVES OF THE STUDY

1. To explore the changing nature of cyberattacks from phishing to deepfakes.
2. To examine the extent to which the current legal framework of India is equipped to deal with modern cyber threats.
3. To analyse gaps in India's enforcement and preparedness for advanced cyber threats.
4. To compare India's cybersecurity ecosystem with the best global models of the US, the EU and China, with an aim of identifying adaptable best practices
5. To suggest a high-priority transition from a largely reactive stance to a proactive future branded cybersecurity governance.

RESEARCH QUESTIONS

1. How has the evolution from traditional cyberattacks to AI-driven threats revealed inadequacies in India's existing legal framework?
2. Does India's existing legal cybersecurity framework adequately address the challenges posed by modern cyberattacks?
3. How can India transform from a reactive to a proactive regulatory model by adopting global best cybersecurity practices?

SCOPE OF THE STUDY

1. **Geographical Boundaries:** The study is not only confined to India but also includes a comparison with leading international cybersecurity models focusing on their legal and technological preparedness against emerging cyber threats.
2. **Subject Scope:** Various types of cyber threats have been covered in the paper. However, the main focus has been given to phishing and deepfakes.
3. **Legal and Institutional Focus:** The paper assesses India's current legal position in tackling cyberattacks. Legislation like the Information Technology Act, 2000 and the Digital Personal Data Protection Act, 2023, etc., has been taken into consideration. The paper also examines the efforts taken by government institutions like CERT-In, NCIIPC, Cyber Crime Units, and NCCC.

4. **Technological Scope:** The paper talks about the different types of technologies behind the cyberattacks, such as AI and machine learning.

METHODOLOGY

1. **Doctrinal Method:** The paper examines India's current legislation dealing with cybersecurity, which includes an in-depth analysis of the Information Technology Act, 2000 and the Digital Personal Data Protection Act, 2023, Bharatiya Nyaya Sanhita, 2023, CERT In 2013 Rules and 2022 Directions and review of institutional efforts taken by the government, such as CERT-In, NCIIPC, Cyber Crime Units, and NCCC.
2. **Comparative Analysis:** International Cybersecurity Models Assessment of advanced international cybersecurity models from countries such as the United States, the European Union, and Singapore
3. **Secondary Data Collection:** Use of published research papers, government reports, legal documents, news articles, and official cybersecurity reports for analysis of trends, statistics, and expert opinions to evaluate the effectiveness of India's cybersecurity laws

LITERATURE REVIEW

Evolution of Cyberthreat in India

In India, the date dates back to the initial stages of internet usage, when the nation first started witnessing the onset of digital fraud. Studies show that one of the earliest notable incidents happened in the mid-1990s, when a group of hackers impersonated AOL employees through instant messaging and email to steal users' passwords and hijack their accounts.¹ These messages used to request login credentials for account verification. This era established the basic framework of phishing, using fraudulent communications to trick users into handing over sensitive information. However, it is not extensively documented in academia.

Another incident was a PayPal scam in the early 2000s, where attackers impersonated PayPal under names like "PayPai", which targets account holders of the widely used internet payment service PayPal, taking advantage of the fact that a capital "i" may be difficult to distinguish from a lower-case "l" in some computer fonts. The scam involves sending PayPal account holders a

¹ J Crumbaugh, "The evolution of phishing attacks", *available at*: <https://www.phishfirewall.com/phishing-playbook-chapters/the-evolution-of-phishing-attacks> (last visited on: 3.04.2026).

notification email claiming that PayPal has “temporarily suspended” their account. Instead of linking to PayPal.com, the site references in the email link to a convincing duplicate of the site at paypai.com, in the hope that the user will enter their PayPal login details, which the owner of paypai.com can then store and use.²

By the early 2010s, phishing had evolved from small-scale fraud to organised digital networks, as in the case of the Jamtara scam in Jharkhand, including the tools and techniques employed by offenders, as well as the socio-economic factors that contribute to their activities. In India, in this case, the scammers used to call people, pretending to be bank officials or telecom executives. They created fabricated situations, like threatening to freeze bank accounts or disable SIM cards, to create a sense of haste. Targets were then bullied into divulging sensitive data, such as bank account numbers, OTPs, and ATM PINs. Over time, the operation expanded, with numerous youths in Jamtara replicating the scheme, earning the district the reputation of India’s “phishing capital.”³

The COVID-19 pandemic marked another surge in phishing attacks, as cybercriminals exploited public fear and the increased reliance on digital services. A study noted that during the Covid 19 pandemic, there was a surge in phishing attacks. Because of the lockdown, people were heavily relying on digital services. Cybercriminals took advantage of this situation by sending malicious emails in the name of charitable purposes to the victims who suffered during the pandemic.⁴

Legal and Institutional Framework in India

A study showed that India has undergone significant development in terms of cybersecurity laws in order to keep pace with emerging digital risks, but is still not fully equipped to tackle emerging cyberthreats like deepfakes. No doubt, the IT Act 2000 acts as the cornerstone of India’s legislation on cybersecurity and has been supported by several subordinate rules aimed at tackling cyber-related risks. Though India has expanded its cybersecurity regime, its effectiveness is significantly affected by enforcement and regulation challenges. Considering the fact that cybercrime may transcend national borders, India should show its willingness to strengthen international partnerships. Such collaboration will help in the exchange of knowledge and coordinated strategies. Moreover, keeping in mind that technological advancements have

² E Gabrilovich and A Gontmakher, “The homograph attack” 45(2) *Communications of the ACM* 128 (2002).

³ R Suman, P.K. Ray and A Sarkar, “A study on cyber financial frauds in the district of Jamtara, Jharkhand” 9(4) *International Journal of Research and Analytical Reviews* 123 (2022).

⁴ M.L. Kumar and P.S. Akram, “Cybersecurity laws in India: Constitutional implications and regulatory challenges” 14(15s) *Journal of Neonatal Surgery* 193 (2025).

transformed simple cyberattacks into sophisticated ones, India should take a shift from a reactive approach to a proactive approach in order to fight against the emerging modern cyberattacks.⁵

Technological aspects of Cyber Threats and AI's Role:

With the emerging rise of new digital advancements, the nature of cyber threats has also transformed. Earlier, they were easy to detect, but modern cyberattacks powered by AI are very difficult to detect due to their adaptive and intelligent nature. The question of whether AI has strengthened or weakened cybersecurity defences has always been debated. Different scholars have different reasonable opinions about it.

Some studies show that AI enhances cybersecurity defences as it relies on several advanced technologies, including machine learning, deep learning and natural language processing (NLP). Machine learning is a subset of AI that is more focused on the use of various self-learning algorithms that derive knowledge from previous data in order to predict outcomes, which helps in preventing cyberattacks by recognising similar patterns in future. Deep learning, also known as neural networks, is a further subset of machine learning. It teaches a computer with many layers of neural networks to recognise complex patterns given in the data, so that it will learn to see, hear and understand things on its own. Natural language processing (NLP) helps AI to read, understand and derive meaning from human languages.⁶

However, some studies say that the technologies, specifically AI, which boost the cybersecurity defences, are the same that can worsen the landscape of cyberattacks, as AI can now generate fake images, videos or audios that seem difficult to identify by a normal person, expanding the attack surface. Computer Using Agents (CUAs), built on platforms like OpenAI Operator, are now able to mimic human behaviours, which raises serious concerns.⁷

This shows how AI can both strengthen or worsen the cybersecurity defences. The focus should be on using AI for strengthening purposes, which requires both technical implementation and strategic oversight.

Gap identification

⁵ M.L. Kumar and P.S. Akram, "Cyber security laws in India – Constitutional implications & gaps" 14(15S) *Journal of Neonatal Surgery* 194 (2025).

⁶ Syracuse University School of Information Studies, "AI in cybersecurity", available at: <https://ischool.syracuse.edu/ai-in-cybersecurity/> (last visited on: 03.04.2026).

⁷ T Seals, "Agentic AI: Cybersecurity's friend or foe?", *TechRadar Pro*, Oct. 31, 2024, available at: <https://www.techradar.com/pro/agentic-ai-cybersecuritys-friend-or-foe> (last visited on: 03.04.2026).

- Although India's cybersecurity legislative framework has undergone significant developments, it is still not fully compatible with dealing with advanced cyberattacks and lacks a reactive and preventive approach.
- Socio-economic factors have always been ignored, as most of the studies focus only on the technical aspects of cybercrime.
- There is a lack of comprehensive research on modern cyberattacks like deepfakes and synthetic media.

UNDERSTANDING CYBER THREATS

A cyber threat is any malicious attempt carried out by either an individual or group through the use of digital tools with the aim of gaining unauthorised access, manipulating data, threatening security, privacy or causing damage. It consists of a wide range of attacks such as phishing, deepfakes, identity theft, and synthetic media.

Classification of cyber threats

1. **Financial Crimes:** In the cyber context, it refers to illegal activities that involve the exploitation of banking networks and digital payment systems, with the help of technological tools like phishing and malware, with the aim of cheating people or stealing sensitive financial details, such as Credit or debit card skimming, UPI fraud, and trading app fraud.
2. **Data Theft:** It is the most common form of cyber threat, which includes illegal extraction of personal data by sending fake links or emails, from databases through malware attacks and hacking. Data theft may include banking information, passwords, photos or any other confidential information.
3. **AI-based manipulation:** As a cyberthreat, AI-based manipulation refers to the modern cyberthreats that are powered by AI to deceive or influence individuals through synthetic media, targeted content, and voice cloning. The fact that AI can analyse behaviour and personal data makes it difficult for normal people to distinguish between real and fake content, such as deepfake videos used for blackmailing, social media bots spreading misinformation, etc.

EVOLUTION OF CYBERCRIME - KEY TRENDS, CYBERSECURITY THREATS, AND MITIGATION STRATEGIES FROM HISTORICAL DATA

According to the study, historically, cybercrimes used to be simple in nature, such as sending fake emails, messages or links to trick people. Such kind of scams are easy to detect because they include suspicious sender addresses, threatening language, poor grammar, etc. But in the last few decades, there has been a huge shift from simple cyberattacks to AI-powered cyberattacks due to rapid digitisation. AI-powered cyberattacks are complex in nature, as AI can analyse behaviour and personal data, which is why they are more convincing.⁸

Earlier, the main target of cybercriminals used to be small enterprises and individuals, as they had limited knowledge about the digital world and technology. This trend is not only technological but also closely linked to underlying socio- economic factors that significantly shape the dynamics of cybercrime in India. In India, differences in digital literacy and educational disparities have a strong impact on both victimisation and offender behaviour. On one hand, a digitally unaware population are disproportionately targeted, making them more vulnerable to cyberattacks as they are not capable of identifying them. On the other hand, disadvantaged groups may find cybercrime an attractive alternative due to its low risk and high reward opportunity. This dual dynamic shows that cybercrime is not about only technology but also requires consideration for socio- economic factors. The case of Jamtara shows how socio-economic factors can foster environments conducive to cyber fraud.

As the technology kept improving, so did the cybercriminals. Technological advancements have spurred the growth of complex cybercrime. The total number of ransomware attacks has increased in contemporary times, where cyber criminals illegitimately encrypt victims' information and then extort substantial ransom payments to provide the decryption key. Not just individuals and corporations have also been impacted by these attacks. Advanced persistent threats (APTs) are another notable trend. These attacks tend to be state-sponsored or state-funded; they are prolonged, stealthy penetrations of complex computer systems to capture confidential information or disrupt operations

The landscape of cybercrime has changed over the years, too. In their earlier days, only small-scale businesses and individuals were targeted by cyber criminals due to insufficient security resources and a lack of awareness. However, with every breakthrough in technology, cyber criminals develop more sophisticated plans, owing to which big organisations, such as the private sector, governments, civil society, and the military, are becoming common targets.

⁸ M Abdullah, M.M. Nawaz, B Saleem, M Zahra, E.B. Ashfaq and Z Muhammad, "Evolution cybercrime-Key trends, cybersecurity threats, and mitigation strategies from historical data" 4 (3) *Analytics* 25 (2025).

The early 1960s were the period when early technological developments birthed to connectivity. The computers during this era were costly, big, and huge. Multiple individuals shared a single computer simultaneously. This sharing of time led to the necessity of avoiding access by unauthorised users to computers and their data. And thus, the idea of data security and hardware security was conceived here. The defence practices during this decade were more focused on physical access control, locked facilities and duties segregation to avoid insider abuse. During the 1970s, ARPANET, the initial version of the internet, was established, which provided hackers everywhere with much to ponder. Some early malware, such as the Creeper and Reaper, were created as well, but they were used as academic exercises, not as actual malware. In 1975, an article called 'The Protection of Information in Computer Systems' was released that provided rules that would serve as the basis of contemporary cybersecurity. The 1980s witnessed the formal establishment of the Internet, and the networks everywhere in the world began implementing the Internet Protocol Suite. The adoption added additional prey and foes to the table. The dictionary attack, which is employed to take advantage of weak or default passwords, was also first deployed during the 1980s. Since the world of the internet was new, the security practices to be implemented were also not yet well understood in this context. In the 1990s, the period is referred to as the virus era. As personal computers became common, they enabled even new users to gain access to ready-made hacking tools without understanding coding. No doubt, many of these attacks were done for experimentation or curiosity, but they highlighted growing risks of cyber threats for technological companies, promoting the adoption of antivirus software, mandatory updates, etc. The 2000s saw the growth of online banking, and reliance on digital services increased, which created opportunities for cybercriminals to commit financial fraud. During the 2010s, there was further evolution of cyberattacks with the emergence of state-sponsored attacks, backed by the government, regarded as the most sophisticated form of cyberattacks. These cyberattacks include a high-skilled hacking group that used to target tech giants and global organisations in order to steal information. The 2020s saw a transition from traditional cyberattacks such as phishing to advanced cyberattacks like deepfakes, which are more difficult to detect as they are powered by AI. During this time, cybercriminals started to use fake but realistic audio and video to pretend to be something else and trick people into sending money or revealing their personal information. A systematic review also brings forth continuing challenges such as under-reporting of cybercrime, lack of adoption of sophisticated detection solutions, and incorporation of psychological, social, and technical factors in research designs. Cumulatively, these findings underscore that the 2020s require a multidisciplinary response leveraging AI-driven detection, regulatory reforms, public awareness, and international

cooperation to effectively counteract evolving cybercrime threats. Talking about the present context, as we are entering a more globalised world each passing day, the threats and dangers of cybersecurity are growing as well, like deepfakes. Emerging technological innovations such as 5G, quantum machines, IoT appliances, and cloud computing services have expanded the attack surface. Therefore, in response to these developments, strong and proactive countermeasures and cautionary steps must be applied.

PHISHING

The term “Phishing” has drawn its resemblance from the word “fishing”, in which one uses an insect as a lure to catch a fish. The moment the fish bites the insect, it gets hooked to the bait. Similarly, phishing is being used. In the digital world, to defraud the victims by luring them into a bait. The forms of bait could take place in the form of a phishing email with the URL of a phishing website, or smishing, a type of phishing scam that takes place through SMS, or vishing / voice phishing, another type of phishing scam that takes place via phone calls. By using all these methods, the phisher tricks the users into revealing personal information such as login credentials, Credit Card CVV, biometrics, details of Aadhaar, Gmail IDs, or any other confidential information that seems valuable to him.⁹ Upon falling for the phisher’s trick, the victim enters their confidential details on this phoney website, the phisher then steals this information and misuses it to transfer the money of the victim.

According to studies, there is a high possibility that the phishers sell the stolen data, such as login credentials or any other sensitive information, on the dark web community in exchange for a large amount of money. The community then initiates further such attacks through these compromised machines or servers. The phishers may also equip themselves through the Dark Web, further helping them with these phishing attacks and expanding their operations. These phishers often inculcate more than one method or technique to initiate more and more such attacks and trick victims through emails, SMSs, etc., to click on fake URLs. Once successful, malware is discreetly downloaded into their machines, and thus, complete, uninterrupted, and discreet access to these networks and machines is with the phisher through additional system Administrator Accounts.¹⁰

There was a surge in phishing email attacks during COVID-19. The Economic Times reported that COVID-19-related phishing attacks grew by 667% in February 2020. Phishing attacks were

⁹ Jaya, “Machine learning based phishing website detection” (2023) (Unpublished doctoral thesis, Jawaharlal Nehru University).

¹⁰ A Oest *et.al.*, “Inside a phisher’s mind: Understanding anti-phishing ecosystem through phishing kit analysis”, in *2018 APWG Symposium on Electronic Crime Research (eCrime)* 1 (2018).

happening in the name of donations for COVID-19, help for health workers and free online meeting software. A report publicised by APWG in its 1st Quarter, 2020, stated that some of the email phishing attacks were carried out by sending fake Zoom video-conferencing meeting notifications to people.¹¹ The report by APWG in its 3rd Quarter, 2022 Phishing Activity Trend Report highlighted that the number of unique Phishing Websites detected in the months of July-Sep 2022 was 425,112 (July 2022), 430,141 (Aug 2022), and 415,630 (Sep 2022).¹²

DEEPPFAKES

Deepfake is one of the most advanced cybercrimes that uses artificial intelligence (AI), which allows the user to mimic the synthetically enhanced video of a real person, providing images and audio-visual to make the user speak or behave in a manner that is not realistic. As deepfakes are based on machine learning, data and information of a person can be easily processed by algorithms and systems, thereby generating any body part on screen, including face, body and other visual characteristics, which appears to be real but is not true. There are three main groups that have been identified by the researchers into which deepfake content falls. The first is Face Swap, replaces a person's face in a video or image with someone else's, creating a seamless and natural looking transformation; the second is Lip- sync, manipulates a person's lip movements to match new or altered audio track, making it appear that they are speaking words they never said; and the third is puppet- master, manipulates the facial expressions of one input at a time. The person in the source video controls the facial expression and movements of the person in the target video.¹³

Deepfakes are a subdivision of deep learning, which is a realm of artificial intelligence that depends on neural networks for its operation. In the realm of deep learning, typically a combination of input and output layers is used to operate through neural networks. The system works in gears with the two main components under the framework, named the generator and the discriminator. Both these components are responsible for the algorithm in use. These two bodies help the system to differentiate between synthetic and authentic data, wherein the generator helps in mimicking real content through artificial outputs, and the discriminator helps in identifying the features that point out fabricated material and also helps check authenticity. As these components function and interact with each other, the sync and specific functions allow the generator to continuously receive feedback from the discriminator, which identifies authentic

¹¹ Anti-Phishing Working Group, "Phishing activity trends report: 1st quarter 2020" (May 11, 2020), *available at*: https://docs.apwg.org/reports/apwg_trends_report_q1_2020.pdf (last visited on: 03.04.2026).

¹² Ibid.

¹³ Ibid.

traits detected by it and allows fake material to become more realistic and standardises it with genuine examples. As the distinction between synthetic and authentic images grows smaller, so does the significance of the input that the model assigns to itself, utilising these measures as a metric for performance accuracy. The elaborative process shapes the foundational base for deepfakes and is called Generative Adversarial Network (GAN), which possesses the capacity to generate and reproduce data patterns to generate deepfakes. The discriminator here plays the most important role in keeping generating feedback, which helps the model to correct itself wherever it lacks authenticity. Without the feedback in place, the model would fail and would not be able to correct itself.¹⁴

INDIA'S LEGAL AND INSTITUTIONAL FRAMEWORK ON CYBERSECURITY

India's legal and institutional framework addresses cybersecurity in India by providing privacy and securing critical infrastructure. India's legal, institutional and regulatory framework has expanded steadily, shaped by several laws, statutes, rules, policies and institutional mechanisms. All these instruments aim to define cybercrimes in order to protect personal information, online transactions and overall ensure a safe digital environment for individuals, corporations and the government.

Legal Framework

The foundational legislative framework that deals with cybersecurity in India is the Information Technology Act, 2000, considered the cornerstone of the country's digital legal architecture. It was enacted when the trend of electronic communication and e-commerce was emerging globally. The Act prevents cybercrimes, regulates electronic transactions and legitimises digital signatures. A wide range of offences and responsibilities has been addressed by the Information Technology Act, 2000. Section 43 of the IT Act, 2000 punishes the offender for data theft, identity theft, hacking a computer system, damaging a computer system or computer network, or causing denial of service. Section 43A of the IT Act, 2000 provides compensation to the affected person if a body corporate viz. any company and includes a firm, sole proprietorship or other association of individuals engaged in commercial or professional activities. Every so often, the generator may generate such refined data that the discriminator fails to tell it apart from the actual and real data, creating a loop wherein both parties oscillate around like a pendulum,

¹⁴ J Gurumurthy, "In the pursuance of a robust legal framework to address deepfake harms: An analysis of the Indian legal discourse" 20(1) *Indian Journal of Law and Technology* (2025).

forming an uncertainty. India's Framework on cybersecurity helps address this issue by providing privacy and securing critical infrastructure.¹⁵

Also, the Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021, notified under the Information Technology Act, 2000, establish a new framework for regulating social media companies, online intermediaries, OTT platforms and digital news publishers, to ensure accountability in relation to their activities. These rules make the "safe harbour" protection under section 79 of the IT Act, 2000 contingent on the intermediary exercising due diligence, including publishing user-agreement policies, responding to complaints within 24 hours and taking down unlawful content within 72 hours of a request by a court or government authority.¹⁶ The rules also classify online platforms as Social Media Intermediaries (SSMIs), platform with over 5 million users in India. SSMIs are required to appoint a chief complaint officer, a nodal contact officer, and a resident grievance officer; store user data for 180 days and publish monthly transparency reports.¹⁷ The most controversial aspect of the rules is the traceability requirement, which mandates that encrypted messaging platforms like WhatsApp must make the first originator of the message identifiable for investigations involving certain types of offences.¹⁸ Scholars argue that traceability undermined end to end encryption and disproportionately infringes on the right to privacy recognized in *Justice K.S. Puttaswamy v. Union of India*.¹⁹ For OTT platforms, The Rules implement a three-tier regulatory arrangement whereby there is a level of self-regulation by the platforms, a level of self-regulation by the industry body, and a level of final determination by a government committee, while digital news publishers must follow the Press Council and Cable Television Programme codes.²⁰ While the Rules were designed to address illegal content and misinformation by enhancing accountability, data shows that unclear obligations to remove certain content and state-controlled oversight will likely lead to censorship and create a chilling effect on free speech.²¹

An amendment came in 2008 in the IT Act 2000, which introduced 9 chapters and 117 sections, which updated the law to tackle modern cybercrimes, reinforce security and data protection standards. It also supported the use of digital signatures, strengthened forensic processes and

¹⁵ M Gupta and A Gupta, "Cyber security legal framework in India - Overlaps, problems and challenges" 12(1) *Journal of Business Management & Information Systems* (2025).

¹⁶ The Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021, r. 3.

¹⁷ Ibid at r. 4.

¹⁸ Ibid at r. 4(2).

¹⁹ (2017) 10 SCC 1.

²⁰ Supra note 16 at pt. III, rr. 8-13.

²¹ G Bhatia, "The Indian Intermediary Guidelines: A constitutional analysis", *Indian Constitutional Law and Philosophy*, Feb. 26, 2021, available at: <https://indconlawphil.wordpress.com/2021/02/26/the-indian-intermediary-guidelines-a-constitutional-analysis/> (last visited on: 03.04.2026).

cybersecurity investigation and acknowledged the necessity of protecting personal data from cyberthreats. Also, the mandate imposed on organisations and intermediaries to notify CERT-In of cybersecurity incidents involves a more structured approach to strengthen the responsive cybersecurity framework.

India has strengthened its legal framework through the Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Data or Information) Rules, 2011, also known as Privacy rules, which were enacted as an accessory to the IT Act, 2000. They recognised certain categories of information, such as biometric data, financial details, passwords, and medical data, as Sensitive Personal Data or Information (SPDI). It imposed compliance requirements on intermediaries and other such entities handling such data. Companies and intermediaries were required to obtain consent from the user, ensuring transparency regarding data usage and implementation of reasonable security practices to prevent unauthorised access. The rules also mandated the establishment of grievance redressal mechanisms and obligated the publication of privacy policies. Later, influenced by global data protection frameworks like the EU'S GDPR, India enacted a comprehensive personal data protection law on August 11, 2023. This law aims to create a secure and accountable data processing environment by allowing individuals to access, correct and withdraw consent for the processing of their personal information. The Act also imposes reasonable yet strict restrictions and obligations upon the Data Fiduciaries (organisations dealing with data) in relation to the regulations and limitations of their power over personal data, and most importantly, to immediately notify any breach to their stakeholders, ultimately reinforcing compliance maintenance and security safeguards. The issue of the protection of children's data has also been taken into account while making the DPDP Act, as the Act ensures that minors are not subject to targeted advertising and data profiling of minors. To oversee and ensure proper and efficient implementation of the DPDP Act, a quasi-judicial body has been set up called the Data Protection Board (DPB) by virtue of the Act. The nature of the Board being that of a quasi-judicial one, it is empowered to not only investigate any violations but also impose penalties on such violators as required and prescribed. The newly enacted DPDP, as mentioned above, is not one of a limited nature offering limited regulations, but is an overarching act that helps in almost every aspect. The act ranges from establishing a safe and secure environment for encouraging business, boosting the economy, by adopting measures for accountability, trust, and compliance. It also addresses national security concerns relating to cyber-attacks and empowers the state and central government to process data for national security. The impact from these establishing and first time legislations cyber security awareness has reached a new level in India and key legislations such the penal code of India now

Bhartiya Nyaya Sanhita, 2023 (BNS) under section 2(39) of the act to increase compatibility and application of the IT act mentions that words and meanings in relation with electronic and digital means is inculcated same as it is in the IT act, thus helping in deterrence and detection crimes through such means. These clarifications and inclusions now help ensure that the core and traditional criminal law is not unknown to the modern technological upgrades, but complements the specialised IT Act 2000.

In 2013, a subordinate legislation aiming to improve cybersecurity laws was enacted, known as the Computer Emergency Response Team (CERT-In) Rules. Through this act, CERT-In was designated as the nodal agency for monitoring and reporting any incident, all the while providing quick response to these incidents. It is empowered to collect data anytime an incident takes place, issue advisories and also seek information from intermediaries, data centres, and service providers to prevent any threats in a timely manner. To add more safeguards and checks, these rules were updated in 2022, which added more obligations such as incident reporting within six hours, retaining data of users for a maximum of 180 days, VPN firms, virtual assistance and much more. These guidelines are foundational grounds for a step towards proactive surveillance and accountability, all the while matching itself with global standards.

A major shift comes with the IT (Intermediary Guidelines & Digital Media Ethics Code) Amendment Rules, 2026. The new rules come into effect from 20 February, 2026. This amendment makes a turning point in India's legal framework governing cybercrime, as it is the first framework to explain the concept of "Synthetically Generated Information" (SGI) explicitly under Rule 2(1)(wa). It defines SGI as AI-generated or AI-altered content which appears to be real, and can mislead users into believing it is authentic, like deepfakes, AI-generated realistic images, and voice cloning. The amendment mainly focuses on regulating AI-generated content. Not every AI-generated content is SGI, like basic editing, which includes noise reduction, video compression, adding subtitles, and normal AI assistance for creating PowerPoint slides, drafting samples. Therefore, this amendment differentiates between content that SGI and content that does not.²²

Key features of the IT (Intermediary Guidelines & Digital Media Ethics Code) Amendment Rules, 2026:

²² Prime Infoserv, "AI & Deepfake Law in India: IT Rules 2026 Amendment Tightens Control Over AI-Generated Content", *Prime Infoserv*, Feb. 11, 2026, available at: <https://primeinfoserv.com/blog-ai-deepfake-law-india-it-rules-2026-amendment/> (last visited on: 03.04.2026).

1. Reduction in response time:

- Intermediaries have to take down the AI-generated content within 3 hours upon receiving court or government orders. Earlier, the time period was 36 hours.
- The timeline for resolving general complaints has been reduced from 15 days to 7 days.
- The timeline for removal of morphed intimate images/ nudity has also been reduced from 24 hours to 2 hours.

2. Mandatory labelling of SGI content:

- Under Rule 3(3), the intermediaries have a mandatory duty to systematically label the SGI content on their platform.
- It must display permanent and unique metadata.
- For deepfake videos, the label must occupy 10% of the screen area and must remain permanently visible.
- For audio deepfakes, mandatory audible disclosure during the opening seconds enables viewers to evaluate the credibility of the content.

3. SSMI (Significant Social Media Intermediaries) requirements:

- Intermediaries with more than 5 million users are considered SSMI.
- SSIMIs are required to deploy AI tools or human review to verify user declarations. The new rule 3(1A) gives protection to intermediaries if they remove synthetic content honestly, either through reasonable efforts or acting on user complaints.

4. Mandatory warning for using AI tools:

- If a platform is providing AI tools to use, then it must also inform the users about the legal penalties for misusing the same
- They must regularly show warnings at sign-up and periodically to ensure continued awareness.

Institutional Framework:

With all these legislations in place, it is clear that cybersecurity governance has taken a step forward; however, these legislations only bind the provisions within the legal boundaries, and thus, the role of implementation further enhances the need for well-structured, accountable, and transparent institutions. India has several dedicated institutions aiming at the prevention of cybercrimes through reporting, policy execution, and mitigation. One of the key institutions in

this context is the Computer Emergency Response Team (CERT-In), which was established in 2004 and plays a crucial role in safeguarding the digital infrastructure of India. It enables India to collaborate with national and international organisations in conducting awareness campaigns. Through its secure and complex network of intelligence and coordination with the cybersecurity framework, the institution has now come to play an indispensable role in maintaining the integrity and security of the nation's digital ecosystem. On 16 January 2014, the National Critical Information Infrastructure Protection Centre (NCIIPC) was officially notified in the Information Technology (Amendment) Act, 2008, under Section 70A. This is another initiative taken by the government. This body is the supreme body for securing Critical Information Infrastructure (CII). NCIIPC is a subset of the National Technical Research Organisation (NTRO) and performs functions such as identifying cybersecurity vulnerabilities across the critical sectors like power, telecom, transport, finance, and government) performing risk assessment and audit, and sharing actionable threat intelligence with stakeholders in the relevant sectors. Through its strong mandate, it focuses on prevention and makes sure that the national sovereignty and functioning are not affected by a cyberattack on critical infrastructure.

Ministry of Electronics and Information Technology (MeitY) has also implemented an institution that helps in real-time analysis and threat prediction called the National Cyber Coordination Centre (NCCC). Its role is to analyse any internet traffic anomalies, send early alerts, and distribute intelligence.

THE DILEMMA OF INEFFICIENT INDIAN LAWS

Though the IT Amendment Rules, 2026 make a significant step forward in India's approach to regulate deepfakes, they are not sufficient on their own. The major drawback is that these amendments are intermediary-centric. It tells platforms to regulate the content, detect harmful AI-generated content, label it and take it down if it misled viewers. However, it does not directly regulate the creation or misuse of deepfakes themselves in India. There is still no specific law defining deepfake offences, no direct punishment for creators, and no specific law targeting its misuse. Therefore, these amendments govern the platforms where harmful AI-generated content is shared rather than the act of creating harmful AI content. These new rules also ignore the ground reality that harm goes beyond platforms. Even if intermediaries are well-regulated, such content can still be shared in private. For example, apps like Telegram and WhatsApp have encryption that makes it difficult to detect such content. Also, people can share such content by sharing files directly from one device to another via Airdrop, Bluetooth or USB services without involving any platform. It helps in regulating the platform, which is only one part of the

problem, but fails to regulate the harm itself. Therefore, the law is not fully effective and continues to remain reactive as it does not address the actual misuse of deepfake technology.

Another major gap left unaddressed by India's cybersecurity legislative framework is jurisdiction. This is mostly true for cross-border crimes, wherein a variation between the jurisdictional gap creates a grey area for cybercriminals, giving them opportunities to escape from liability by operating from within nations or places that are heavily ill-equipped with their cybercrime laws, leading to limited enforceability. Such variability is often seen in these cyber-attacks, mostly prominent for hacking. Hacking often is initiated from countries with outdated cyberlaw, making it difficult for law enforcement to not just apprehend criminals but also help in mutual legal compliance with other countries. Additionally, advancements in hacking have allowed it to start the same attack from not one but several jurisdictions, making it even tougher to trace and locate the origins of the hacker. Thus, such an inconsistency in the legal framework often leads to a lack in the justice system when it comes to a speedy response to such attacks, as law enforcement often is ambiguous or stuck in its own procedures, protocols, and timing.²³ Studies also suggest that India's lack of action against transboundary crimes is not in its framework, but rather in the related issues of its ability to collect, preserve, and admit evidence outside the jurisdiction of Indian boundaries.

Various policy papers have suggested that India is very much dependent upon the Mutual Legal Assistance Treaty (MLAT), an official agreement between two or more nations that allows signatories to exchange evidence and information for purposes of investigation and prosecution of crime, which is unstandardized and slow and results in the loss of important server log evidence.²⁴ To address these issues, scholars have put forth a response model that is in 3 layers. Firstly, at the legal level, India should enter into bilateral data access treaties with significant jurisdictions similar to the U.S. CLOUD Act and implement emergency data preservation orders with the ability to bind foreign intermediaries.²⁵ Secondly, at the institutional level, India should designate the NCCC as the key agency to coordinate CERT In, NCIIPC and the 24/7 Interpol Network for cross-border evidence.²⁶ Third, at the diplomatic level, researchers have suggested

²³ M.T. Hasan, "Cross-border cybercrimes and international law: Challenges in ensuring justice in a digitally connected world" 4(1) *IJRDO Journal of Law and Cyber Crime* 1–7 (2024).

²⁴ Carnegie India, "Cross-border data access for law enforcement in India", *Carnegie Endowment for International Peace*, June 12, 2020, available at: <https://carnegieindia.org/2020/06/12/cross-border-data-access-for-law-enforcement-in-india-pub-81981> (last visited on: 03.04.2026).

²⁵ SSRN, "Issues with enforcing MLATs in cybercrime cases: An empirical study" (2022), available at: <https://ssrn.com> (last visited on: 03.04.2026).

²⁶ CCDCOE (Cooperative Cyber Defence Centre of Excellence), "Mutual legal assistance and improving international cooperation in cybercrime investigations" (2021), available at: <https://ccdcoe.org> (last visited on: 03.04.2026).

that India should adopt the standards of the Budapest Convention to allow for faster admissibility of digital forensic reports and joint investigation teams. Overall, together, these reforms would shift India from a reactive MLAT-based model to a proactive, rapid-response mechanism to collect evidence in a timely manner and enhance cyber deterrence.

Another stage where the enforcement deficit becomes very conspicuous is the investigation stage. Research from the National Law University Delhi 2023 states that nearly 70% of cyber police stations lack trained professionals in digital forensics and that most of the officers are not able to conduct proper electronic evidence.²⁷ Thus, filing of chargesheets after electronic evidence deteriorates, which results in high complaints but very low rates of conviction. According to the Annual Cybercrime Investigation Report, 2024, prepared by the Data Security Council of India, the average length of a cybercrime investigation in India is about 18-36 months, with conviction rates under 2% primarily due to weak handling of evidence, lack of specialist prosecutors and outdated forensic infrastructure.²⁸ Moreover, reports from training this year indicate that many police stations do not have access to even the most basic digital forensic imaging tools, requiring the refusers to be escalated to already overburdened state labs, where again, months of delay are built into the justice system. Collectively, these points toward a systematic problem, i.e. laws are in place, but enforcement capacity is limited. Regulatory ambiguity creates confusion for platforms on compliance, and limitations in investigative capacity mean that very few cybercrimes that are even reported lead to convictions. Thus, the cyber law problem is not about drafting more laws, but rather about establishing clearer, workable regulatory standards and enforcement institutions with technical capacity.

SUGGESTIONS AND CONCLUSION

As India gears up to frame a regulatory response to the evolving complexities of legal disputes in synthetic media and deepfake systems, it can draw inspiration from three international frameworks that offer insightful lessons.

Firstly, the EU Artificial Intelligence Act adopts a risk-based model, wherein AI is classified as permitted or not permitted based upon the classification and also mandates the labelling of AI-generated or manipulated image, audio or video files (for example, deepfakes) as AI-generated. Thus, the need for mandatory upfront labels and disclosure easily tackles the scope for misuse of

²⁷ National Law University Delhi, “Digital forensics capacity gap in Indian policing” (Research working paper, 2023), *available at*: <https://nludelhi.ac.in> (last visited on: 03.04.2026).

²⁸ Data Security Council of India, “Annual cybercrime investigation report” (2024), *available at*: <https://dsci.in> (last visited on: 03.04.2026).

deepfakes at least at a primary level, given that such regulations will only be practised in legal and regulated intermediaries, giving grounds for an informed social media content community and also places a responsibility on makers and distributors of any such content.

Secondly, in the US, such as the DEEPFAKES Accountability Act and various state laws. Its approach is more focused on individual rights and legal remedies. For instance, creating deepfake content without a person's consent, especially in cases of election manipulation and fake pornography, can be illegal. Victims have the right to approach the court directly and claim compensation.

Thirdly, China's Administration of Deep Synthesis of Internet-based Information Services requires that users must verify their real identity before creating or uploading deepfake content, as it makes it easier to trace offenders. The law required platforms to monitor and control their AI systems and required them to audit and report on how their systems work.

Simply put, India's legal framework on cybersecurity needs to undergo several significant changes in order to keep pace with new technological advancements. Nevertheless, India has transformed its cyberlaws, especially the newly introduced IT Amendment Rules 2026, which explicitly address deepfakes and focus on regulating AI-generated content, but continues to remain reactive and insufficient to address the growing risks posed by deepfakes, as it is majorly intermediary-centric. India can take help from different successful international models, such as risk- based classification from the EU, a victim-centric approach from the US, and user identity verification from China. Ultimately, in order to combat modern cyber threats, India requires a multi-dimensional approach and collaborative efforts

WOMEN'S RESERVATION IN LEGISLATURE: A CONSTITUTIONAL, POLITICAL, AND SOCIO-LEGAL ANALYSIS

- Anju Bala*

Abstract

India's Parliament has held thirteen general elections since independence. Not once, across those seven decades, have women crossed fifteen percent of Lok Sabha seats. That gap (stubborn, structural, and politically convenient for those who benefit from it) is what the Nari Shakti Vandan Adhiniyam of 2023 was enacted to close. Whether it will actually do so is a harder question than the near-unanimous vote that passed it might suggest. This paper examines the Constitution (106th Amendment) Act, 2023, through three converging lines of inquiry: its constitutional legitimacy, the political forces that shaped and now constrain it, and the social justice implications of a gender quota that notably omits OBC women from specific protection. The analysis draws on Supreme Court jurisprudence from Kesavananda Bharati to Vijayakumar, Law Commission reports, parliamentary debates, and comparative data from Rwanda, Argentina, Sweden, South Africa, and the United Kingdom. The paper's core argument is that the amendment is constitutionally unassailable but structurally incomplete, its deferred commencement tied to an overdue census, its silence on caste-gender intersection, and its indifference to intra-party gatekeeping together ensure that legal reservation (standing alone) will not deliver the transformative representation it promises.

Keywords: *Women's Reservation, Substantive Equality, Reservation, Delimitation, Gender Justice.*

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INTRODUCTION

Women's political under-representation in India is not a recent problem, nor is it a minor one. Seven decades after independence, the Lok Sabha has never crossed fifteen percent women members.¹ The Constitution of India, in its very Preamble, pledges political justice to every citizen, and Article 14 guarantees equality before law.² Yet these solemn commitments have coexisted with a legislature that is, by any honest measure, overwhelmingly male. The September 2023 passage of the Nari Shakti Vandan Adhiniyam (the Constitution 106th Amendment Act) was Parliament's belated acknowledgement of this contradiction.

The Amendment inserts Articles 330-A and 332-A into the Constitution, mandating one-third reservation of seats for women in the Lok Sabha, State Legislative Assemblies, and the Delhi Assembly. Presidential assent followed on 28 September 2023.³ The vote margins (454 to 2 in the Lok Sabha and 214 to 0 in the Rajya Sabha) suggested near-unanimity. But the fine print told a different story. The reservation will not take effect until after a national census is conducted and a delimitation exercise is completed on the basis of those figures.⁴ India's census, due in 2021, remains unconduct. The mid-2030s is the most optimistic realistic estimate for implementation.

This paper analyses the Amendment through three overlapping lenses: constitutional validity, political and social impact, and comparative global experience. It argues that the Amendment is constitutionally sound but operationally incomplete. The deferred commencement, the absence of an OBC sub-quota, and the silence on intra-party gatekeeping collectively ensure that the formal legal promise will not, by itself, produce the transformative change it names.

HISTORICAL BACKGROUND OF WOMEN'S RESERVATION IN INDIA

Before Independence: The Long Wait for the Franchise

The demand for women's political representation in India predates independence by decades. A delegation of Indian women appeared before Edwin Montagu in 1917, pressing for the franchise. They were politely ignored.⁵ The All-India Women's Conference, founded in 1927,

¹ IPU Parline Database on National Parliaments, Monthly Ranking of Women in National Parliaments (March 2024), *available at*: ipu.org/parline (last visited on: 10.05.2026).

² Constitution of India, arts. 14 and 15(1).

³ The Constitution (One Hundred and Sixth Amendment) Act, 2023 (22 of 2023) [hereinafter the 106th Amendment].

⁴ Ibid, art. 330A (proviso)

⁵ Geraldine Forbes, *Women in Modern India* 67 (Cambridge University Press, Cambridge, 1996).

kept the pressure on through the Simon Commission, the Round Table Conferences, and successive rounds of colonial constitutional negotiation. The Government of India Act, 1935 eventually extended limited voting rights to propertied women, a partial concession that satisfied no one. The structural question of reserved seats was deferred to the post-independence generation.

The Constituent Assembly's Deliberate Omission

It was not ignorance that led the Constituent Assembly to leave parliamentary reservation out of the Constitution. Hansa Mehta, serving on the Sub-Committee on Fundamental Rights, opposed reserved seats, not because she thought women were adequately represented, but because separate electoral rolls for women would echo the communal logic of the Communal Award that had helped fracture British India.⁶ The Assembly chose formal gender-neutral rights in Articles 14 and 15(1), with Article 15(3) as a permissive safety valve⁷, allowing but not requiring special provisions for women. That choice was defensible in 1949. Whether it remained so across seven more decades is the question the 2023 Amendment finally, if incompletely, answers.

The Panchayati Raj Precedent: 73rd and 74th Amendments (1992)

The Constitution (73rd and 74th Amendment) Acts of 1992 changed the terms of the debate fundamentally.⁸ Articles 243D and 243T mandated one-third reservation for women in Panchayati Raj Institutions and Urban Local Bodies, with sub-quotas for SC/ST women.⁹ Within a decade, over a million women held elected local governance positions, the largest single expansion of women's political participation in any democracy. Female-headed panchayats built more water facilities, invested more in schools, and showed lower rates of petty corruption than male-led counterparts.

The most rigorous evidence came from Chattopadhyay and Duflo, whose randomised natural experiment across West Bengal and Rajasthan found that villages with reserved female pradhans invested significantly more in drinking water infrastructure and girls' education.¹⁰ This research

⁶ Constituent Assembly of India Debates, 781 (December 17, 1948), *available at*: cadindia.clpr.org.in (last visited on: 10.05.2026).

⁷ Constitution of India, art. 15, cl. 3.

⁸ Constitution (Seventy-Third Amendment) Act, 1992 (20 of 1992); The Constitution (Seventy-Fourth Amendment) Act, 1992 (21 of 1992).

⁹ Constitution of India, arts. 243D(3) and 243T(3).

¹⁰ Raghavendra Chattopadhyay and Esther Duflo, "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India" 72 (5) *Econometrica* 1409 (2004). See, Animesh Kumar (ed.), *DEEPENING DECENTRALISATION: An Analysis of Local Governments in India Over the Last Three Decades* (Anagh, 2024) *available at SSRN*: <https://ssrn.com/abstract=5236713> (last visited on: 02.06.2026).

became the empirical foundation of the parliamentary reservation argument for the next fifteen years, and also generated the most durable argument against it: the pati panchayat problem, where husbands effectively ran the show from behind elected wives.

Twenty-Seven Years of Near Misses: 1996-2023

The Women's Reservation Bill was first introduced in the Eleventh Lok Sabha in 1996. It lapsed when the government fell. It was reintroduced in 1998, 1999, 2002, and 2003, each time meeting some combination of coalition arithmetic, OBC sub-quota demands, and parliamentary procedure problems. The most agonising near-miss was March 2010, when the Rajya Sabha passed it 186 to 1.¹¹ Champagne was premature. The Bill sat in the Lok Sabha for thirteen more years, never brought to a vote.

The September 2023 passage came fast, almost suspiciously so. Three days from introduction to Presidential assent.¹² The ruling party had a commanding majority and needed no coalition management. The near-unanimous votes (454 to 2, then 214 to 0) masked the absence of any real parliamentary deliberation on the deferred commencement clause that effectively postpones the Amendment's operation by a decade or more.

CONSTITUTIONAL FRAMEWORK

Articles 14 and 15: The Equality Architecture

Article 14 guarantees equality before the law and equal protection of the law. Courts have long recognised that this does not mean identical treatment regardless of circumstance, it means rational, non-arbitrary differentiation. The Supreme Court's landmark reformulation in *E. P. Royappa v. State of Tamil Nadu*¹³ captured this precisely: Article 14 is a guarantee against arbitrariness. Reserving one-third of parliamentary seats for women is not arbitrary; it corrects for the century-old, documented exclusion of women from formal political power.

Article 15(1) prohibits sex-based discrimination by the State.¹⁴ The apparent paradox dissolves when Article 15(3) is read alongside it. Article 15(3) carves an express exception: nothing in Article 15 prevents special provisions for women. The Supreme Court in *Government of Andhra*

¹¹ Rajya Sabha Debates on March 9, 2010 (Constitution (One Hundred and Eighth Amendment) Bill passed 186 votes to 1), *available at*: rajyasabha.nic.in (last visited on May 10, 2026).

¹² Lok Sabha Debates on September 19, 2023 (Constitution (One Hundred and Twenty-Eighth Amendment) Bill passed 454-2), *available at*: sansad.in (last visited on May 10, 2026); Rajya Sabha Debates on September 21, 2023 (Bill passed 214-0), *available at*: sansad.in (last visited on May 10, 2026).

¹³ AIR 1974 SC 555

¹⁴ Constitution of India, art. 15, cl. 1.

*Pradesh v. P. B. Vijayakumar*¹⁵ held that this exception is enabling in character, not merely protective. It authorises positive discrimination, not just safety-net welfare.

Directive Principles as Interpretive Context

Article 39(a) directs the State to secure equal livelihood rights for men and women; Article 46 directs the promotion of the interests of weaker sections.¹⁶ Neither is independently justiciable; Article 37 is clear on that point.¹⁷ But the Supreme Court has consistently treated DPSPs as shaping the interpretive environment in which fundamental rights are read.

The Dignity Dimension of Article 51A (e)

Fundamental duties attract less constitutional attention than they deserve. Article 51A(e) asks every citizen to renounce practices derogatory to women's dignity.¹⁸ Whether the systematic exclusion of women from legislative power qualifies as such a practice is a question worth asking openly. It does not create a justiciable cause of action. But it frames exclusion itself as constitutionally problematic.

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Legislative Enactment and Key Provisions

Strip away the ceremony and the name, and the 106th Amendment does three concrete things. First, it inserts Article 330A, requiring women to hold at least one-third of all directly elected Lok Sabha seats, with a built-in sub-quota for SC/ST women within that one-third. Second, Article 332A replicates the framework for every State Legislative Assembly. Third, Article 239AA is amended to bring Delhi's Assembly into the same arrangement.¹⁹ Reserved constituencies rotate after each delimitation; the reservation runs for fifteen years from commencement, extendable by Parliament.

Then comes the proviso. Commencement occurs only after: (a) a national census is conducted and its figures published; and (b) a delimitation exercise based on those figures is completed.²⁰ The government's stated rationale, that constituency boundaries cannot fairly be drawn without

¹⁵ (1995) 4 SCC 520 at para. 19.

¹⁶ Constitution of India, arts. 39(a) and 46.

¹⁷ *Ibid.*, art. 37.

¹⁸ *Ibid.*, art. 51A, cl. (e).

¹⁹ *Supra* note 3, arts. 330A (1), 332A and 239AA (as amended). *See*, Animesh Kumar, "Nari Shakti Vandan Adhiniyam 2023 An Analysis" 11 (1) *Kanoon Review* 42 (2024).

²⁰ *Supra* note 3, art. 330A (proviso).

current population data, has technical merit. What the Amendment conspicuously omits is any timetable enforcing that sequence. The census due in 2021 remains unconduct. No legislation sets a deadline.

Constitutional Validity and Basic Structure

Can the 106th Amendment survive a basic structure challenge? Almost certainly yes. The doctrine established in *Kesavananda Bharati v. State of Kerala*²¹ protects the constitutional core from destruction by Parliament. Substantive equality is part of that core. An amendment that advances substantive equality, by correcting the structural underrepresentation of half the population, does not abrogate the basic structure; it reinforces it.

The Deferred Commencement as Constitutional Risk

Grover and Tandon put it bluntly: the deferred commencement converts a constitutional right into an executive-controlled aspiration.²² This is not a minor regulatory scheme; it is a constitutional amendment affecting the composition of Parliament itself. Deferring its operation indefinitely to a census whose schedule is entirely at the government's discretion is, at minimum, a structural anomaly that future parliamentary reform must address.

The OBC Sub-Reservation Lacuna

OBC women are the most politically significant omission from the 2023 Amendment. Conservative estimates put them at over 40 percent of India's female population. They carry the double burden of gender subordination and caste disadvantage.²³ The Amendment carves a sub-quota for SC/ST women but does nothing for OBC women, who must now compete within the general women's category against candidates with very different levels of access to party networks, money, and social capital.

JUDICIAL PERSPECTIVE AND IMPORTANT CASE LAWS

*Indra Sawhney v. Union of India*²⁴

²¹ AIR 1973 SC 1461

²² Vrinda Grover and Tripti Tandon, "Nari Shakti Vandan Adhiniyam: Is the Devil in the Detail?" 58 (41) *Economic and Political Weekly* 13 (2023).

²³ Nivedita Menon, "The Women's Reservation Bill and the Illusion of Adequate Representation" 59(5) *Economic and Political Weekly* 22 (2024).

²⁴ AIR 1993 SC 477

Any serious engagement with Indian reservation law begins here. The nine-judge bench's holding on OBC employment reservation under Article 16(4) established principles that still govern: backwardness is social and educational, not merely economic; the fifty percent ceiling exists, and courts will enforce it; reservation is a tool of substantive equality, not permanent entitlement.

*Government of Andhra Pradesh v. P. B. Vijayakumar*²⁵

This is the case that most directly validates the 2023 Amendment's constitutional foundation. The Supreme Court upheld a rule reserving thirty percent of State service posts for women under Article 15(3) and characterised that provision as enabling, going beyond protection to authorise affirmative discrimination.

*Kishok Chandra Pati Tripathi v. Hari Shankar Srivastava*²⁶

Decided before the 73rd Amendment existed, this Allahabad High Court ruling upheld panchayat seat reservation for women, found it within State legislative competence, and rejected an Article 14 challenge. It is an early data point in a consistent line of authority.

*Pramati Educational Trust v. Union of India*²⁷

In *Pramati*, the Supreme Court reaffirmed that substantive equality, located at the intersection of Articles 14, 15, and 21, forms part of the basic structure. Invoking the basic structure against gender reservation is, in doctrinal terms, self-defeating.

*State of Karnataka v. Smt. Sumathi*²⁸

The Karnataka High Court confronted the rotation challenge directly. Rotating reserved panchayat chairperson seats across constituencies does not violate Article 14 or any freedom under Article 19(1)(a). Since the 2023 Amendment uses the same rotation logic, Sumathi forecloses that line of attack before it starts. Across this body of case law, a consistent thread runs,²⁹ every court that has reviewed women's legislative reservation has upheld it.

POLITICAL AND SOCIAL IMPACT

Why Presence Matters

²⁵ (1995) 4 SCC 520

²⁶ AIR 1978 All 154

²⁷ (2014) 8 SCC 1

²⁸ ILR 2007 Kar. 4551

²⁹ M.P. Jain, *Indian Constitutional Law* 892-897 (LexisNexis, New Delhi, 8th edn., 2018).

Anne Phillips spent the better part of *The Politics of Presence*³⁰ arguing something that sounds obvious but is harder to establish than it looks: a decision-making body cannot reliably represent experiences it does not contain. Women's safety on public transport, maternal health funding, domestic violence law reform, and the gender pay gap are under-resourced not because male legislators are heartless, but because the people who most directly experience these problems are not in the room in numbers sufficient to make them politically costly to ignore.

Chattopadhyay and Duflo tracked villages over multiple electoral cycles and found that female-reserved panchayats invested measurably more in drinking water and girls' schooling.³¹ The mechanism is the same at any level: people in power tend to prioritise what affects them.

The Proxy Problem, Honestly Assessed

The proxy argument, that elected women will be fronts for their husbands or male relatives, deserves honest engagement rather than dismissal.³² It is documented in Rajasthan, Bihar, and Uttar Pradesh. What the evidence does not establish is that the proxy phenomenon is permanent. Research tracking the same communities across multiple panchayat terms finds that the dynamic weakens.

Intersectionality and Compound Exclusion

Crenshaw's intersectionality framework,³³ developed in a very different national context, has a precise application here. The barriers facing a Dalit woman from rural Bihar are not simply the barriers facing any woman plus the barriers facing any Dalit, they are a qualitatively distinct compound disadvantage that neither a gender quota alone nor a caste quota alone can address.

ARGUMENTS IN FAVOUR OF WOMEN'S RESERVATION

The democratic legitimacy argument starts with arithmetic. Women are 48.4 percent of the population and roughly the same share of the electorate.³⁴ A legislature where they hold 13.6 percent of seats represents certain people far better than others. First-past-the-post systems disadvantage candidates who lack inherited constituency networks, established party relationships, and independent campaign financing.

³⁰ Anne Phillips, *The Politics of Presence* 5-8 (Oxford University Press, Oxford, 1995).

³¹ Supra note 10 at pp. 1435-1438

³² Bina Agarwal, "Participatory Exclusions, Community Forestry, and Gender" 29(10) *World Development* 1623 (2001) at pp. 1627-1629.

³³ Kimberle Crenshaw, *Demarginalizing the Intersection of Race and Sex* 139 (University of Chicago Legal Forum, 1989)

³⁴ Election Commission of India, "Statistical Report on General Elections to the Eighteenth Lok Sabha" p. 47 (2024) (Table 7).

The substantive equality argument is embedded in the Constitution's own text. Article 15(3) exists precisely because the founders understood that formal equal treatment of structurally unequal parties produces unequal outcomes.

The role model effect is documented. Beaman and colleagues found that exposure to female pradhan leadership, over a single electoral cycle, measurably reduced gender bias in adolescent girls' career aspirations.³⁵ Applied to Parliament, India's most visible political institution, the effect could shift expectations for an entire generation.

Finally, India's international obligations are not ceremonial. CEDAW's³⁶ and the Beijing Platform's 30 percent target³⁷ were ratified with full knowledge of what they required. The 2023 Amendment is the delayed fulfilment of commitments India has been making on international platforms since 1993.

CRITICISMS AND CHALLENGES

Begin with the obvious: the amendment does not apply yet, and may not for a decade.³⁸ The government that passed it will face at least two more general elections before any reserved seats exist under it.

The OBC sub-quota demand is a principled objection, not a procedural technicality.³⁹ A quota that operates as though all women face identical barriers is a fiction. That outcome is upper-caste advantage wearing gender equity's clothes.

The dynasty risk is underappreciated.⁴⁰ Indian political parties have no statutory internal democracy requirements. Documented precedent from panchayat reservation shows parties regularly fielding wives, daughters, and daughters-in-law of existing male politicians.

³⁵ Lori Beaman, Raghavendra Chattopadhyay, Esther Duflo, Rohini Pande and Petia Topalova, "Powerful Women: Does Exposure Reduce Bias?" 124(4) *Quarterly Journal of Economics* 1497 (2009) at pp. 1528-1531.

³⁶ Convention on the Elimination of All Forms of Discrimination Against Women, 1979, art. 7, 1249 UNTS 13 [hereinafter CEDAW].

³⁷ UN General Assembly, Beijing Platform for Action, Fourth World Conference on Women, UN Doc. A/CONF.177/20 (September 15, 1995) para. 190.

³⁸ *Supra* note 3, art. 330A (proviso); *supra* note 22 at pp. 14-16.

³⁹ Rajya Sabha Debates on March 9, 2010 (statements of Mulayam Singh Yadav, SP, and Sharad Yadav, JD(U), demanding OBC sub-reservation as a condition of support for the Bill), *available at*: rajyasabha.nic.in (last visited on May 10, 2026).

⁴⁰ Shirin M. Rai, "Political Parties and Women's Political Representation in India" in Bidyut Chakrabarty (ed.), *Litigating Democracy in South Asia* p. 145 at p. 158 (Routledge, London, 2021).

The rotation mechanism creates a structural disincentive.⁴¹ Long-term political careers require geographical continuity. If a woman's constituency reverts to open competition after delimitation, the entire foundation of her political career disappears overnight.

COMPARATIVE ANALYSIS WITH OTHER COUNTRIES

Rwanda

Rwanda's legislature is 61.3 percent female, the highest proportion in any national legislature globally.^{42,43} The 2003 Constitution mandated at least 30 percent women in all decision-making organs, implemented through reserved seats and closed party lists. The lesson is straightforward: rapid and legally mandated transformation of legislative gender composition is possible when political will exists.

Sweden

Sweden reaches 46.4 percent women in the Riksdag without any constitutional reservation, through voluntary party zipper lists.⁴⁴ Sweden's result took sixty years of sustained feminist organising, a proportional representation system, and a political culture already well above the global median on gender equality. India has none of these preconditions in sufficient measure.

South Africa, Argentina, and United Kingdom

South Africa reaches 45.7 percent women through the ANC's voluntary internal parity policy combined with proportional representation.⁴⁵ Argentina's 1991 Ley de Cupos required 30 percent women on candidate lists; a 2017 parity law pushed representation to 45 percent by 2023.⁴⁶ Labour's All-Women Shortlists in the UK, authorised by the Sex Discrimination (Election Candidates) Act 2002,⁴⁷ doubled female MPs in a single election. Across these examples, gender parity does not happen without deliberate institutional design.

STATISTICAL AND DATA-BASED ANALYSIS

⁴¹ Mrinal Satish, "Women's Reservation and Constitutional Discourse" 46(32) Economic and Political Weekly 74 (2011) at p. 79.

⁴² The Constitution of Rwanda, art. 9 (2003, as amended 2015).

⁴³ IPU Parline, Rwanda: Chambre des Deputes, *available at*: ipu.org/parline/RW (last visited on: 31.03.2024).

⁴⁴ IPU Parline, Sweden: Riksdag, *available at*: ipu.org/parline/SE (last visited on: March 31, 2024).

⁴⁵ IPU Parline, South Africa: National Assembly, *available at*: ipu.org/parline/ZA (last visited on: 31.03. 2024).

⁴⁶ Ley No. 24.012 (Argentina) (November 3, 1991); Ley No. 27.636 (Argentina) (June 24, 2021).

⁴⁷ Sex Discrimination (Election Candidates) Act 2002 (c. 2) (UK).

In 1952, 22 women were elected to the First Lok Sabha, 4.4 percent. The Seventeenth Lok Sabha (2019) reached a historic peak of 78 women, 14.4 percent. The Eighteenth (2024) came in at 74, or 13.6 percent.⁴⁸ At the average rate of increase across seven decades, India reaches 33 percent sometime around 2060. India sits at 148th of 193 countries in the IPU's March 2024 rankings.⁴⁹ The global average stands at 26.9 percent. India is 13 percentage points below it.

Across all State Legislative Assemblies, the average women's representation sat around 9.4 percent in 2024.⁵⁰ India ranked 108th on the UNDP Gender Inequality Index in 2023-24.⁵¹ Legislatures with more women invest more in the public goods that reduce gender inequality.

SUGGESTIONS FOR REFORMS

Firstly, fix the commencement problem with a statutory deadline.⁵² Parliament should prescribe that the Presidential commencement order must be issued within five years of assent, by September 2028.

Secondly, legislate OBC sub-reservation. A constitutional amendment or parliamentary statute under Article 15(3) should create a specific sub-quota for OBC women within the women's one-third.

Thirdly, amend the Representation of the People Act.⁵³ Parties should field a minimum percentage of women in competitive constituencies, starting at 20 percent and escalating to 33 percent across two electoral cycles.

Fourthly, enact anti-proxy legislation. A statutory provision should criminalise the effective control of an elected woman's legislative functions by a male relative or party official.

Fifthly, build political capacity systematically.⁵⁴ A properly funded National Institute for Democratic Participation should provide legislative skills training and sustained political mentoring for women at every level.

⁴⁸ Election Commission of India, "Statistical Report on General Elections to the First Lok Sabha, 1951-52" p. 23 (1952) (Table 3); Election Commission of India, "Statistical Report on General Elections to the Seventeenth Lok Sabha" p. 31 (2019) (Table 7); Election Commission of India, *supra* note 34.

⁴⁹ *Supra* note 1 (global average 26.9%; Bangladesh 21.5%; Nepal 34.4%).

⁵⁰ National Commission for Women, "Women in Politics and Decision Making in India: Beyond Numbers" pp. 34-37 (NCW, New Delhi, 2018).

⁵¹ UNDP, Human Development Report 2023-24 pp. 284-287 (UNDP, New York, 2024).

⁵² Law Commission of India, "Electoral Reforms" Report No. 255 (Law Commission of India, New Delhi, 2015) paras. 4.1-4.9.

⁵³ The Representation of the People Act, 1951 (43 of 1951).

Sixthly, study electoral system reform.⁵⁵ Every country with above 40 per cent women's parliamentary representation uses proportional representation in some form. The Law Commission should study mixed-member proportional systems within three years.

CONCLUSION

The Nari Shakti Vandan Adhiniyam is real legislation, not theatre. It amends the Constitution. It creates a legal entitlement to one-third women's representation that did not exist before September 2023. Courts will uphold it. Its basic structure credentials are sound. Seventy-five years of slow, uneven progress toward gender parity in Indian legislatures now has a constitutional floor, and that matters.

But the conditions attached to that floor are significant enough to warrant something more than celebration. An amendment whose commencement depends on a census the government has not scheduled, implemented through a party system the amendment does not regulate, benefiting a category of 'women' defined without reference to the caste hierarchies that make the category internally unequal, this is a foundation on which something transformative might be built. It is not transformation by itself.

The comparative record is instructive without being prescriptive. Rwanda achieved 61 percent through constitutional mandate. Sweden achieved 46 percent through voluntary party action. Argentina moved from 5.8 percent to 45 percent through escalating statutory candidate quotas. What these examples share is a common recognition that gender parity in legislatures does not happen without deliberate institutional design. India has the constitutional design. What it still lacks is the institutional architecture to make that design operational.

The Preamble promised political justice.⁵⁶ The equality guarantees in Articles 14 and 15 and the international commitments in CEDAW and Beijing are specific about what that promise requires.⁵⁷ The 2023 Amendment moved the constitutional text closer to its own promises. The remaining distance is not rhetorical, it is institutional, political, and structural. Closing it is the work that remains.

⁵⁴ Rohini Pande and Deanna Ford, "Gender Quotas and Female Leadership: A Review" pp. 14-17 (World Bank, Washington D.C., 2011). *See also*, Animesh Kumar, "Gender Equality and Local Democracy: An Evaluation of Women Representation and Participation in Panchayati Raj Institution in India" in Dr. Taruna, & Dr Arpit Shailesh (eds.), Building Atma Nirbhar Bharat 41 (1st ed.), Weser Books (2022), *available at SSRN*: <https://ssrn.com/abstract=5022929> (last visited on: 08.06.2026).

⁵⁵ Rekha Saxena, "Women's Reservation and the Politics of Presence" 5 (2) *Studies in Indian Politics* 210 (2012)

⁵⁶ The Constitution of India, Preamble.

⁵⁷ The Constitution of India, arts. 14 and 15(3); *supra* note 36, art. 7; *supra* note 37.